Computed-Torque Control for Robotic-Assisted Tele-Echography Based on Perceived Stiffness Estimation

Luís Santos and Rui Cortesão

***Abstract*— This paper discusses a new control approach for robotic-assisted tele-echography. The control architecture follows a hierarchical approach, where explicit Cartesian force control arises as the primary task while orientation control is designed in the null space. The robot dynamics is driven by a 3-D time-of-flight camera and a force sensor. Based on depth camera and force data, contact stiffness is anticipated, allowing control adaptation before contact. This approach is adequate for tele-echographic tasks since it reduces robot dynamics before contact, enabling a smooth transition from free space to contact and vice versa. In contact, the environment stiffness is estimated online using force data and the manipulator inertial properties. A strong correlation between the stiffness perceived by the controller and the effective mass exists, being this correlation used in the estimation algorithm to improve force control performance. A set of experiments assess the control architecture, highlighting the relation between perceived stiffness and robot effective mass, including also clinical validation.**

***Note to Practitioners*—The teleoperation control architecture described in this paper relies on computed-torque techniques that take into account robot dynamics, contact stiffness estimation, task-space formulation, and null-space design. Our application is on the medical field, where a physician telecontrols a robot to obtain ultrasound images from a patient. The robot exhibits compliant behaviors in Cartesian space (for safety reasons) managed by the task-space formulation, and stiffer behaviors in orientation control managed by null-space design. This approach allows accurate orientation control with compliant positioning which is adequate for this type of teleoperation tasks. Due to free space and contact interactions, the proposed control approach anticipates the contact stiffness relying on vision sensing, adapting the control gains accordingly. While in contact, the perceived stiffness has a strong correlation with the effective mass, which is used together with force data for its estimation. This strategy allows slowing down robot dynamics toward contact, guaranteeing a smooth behavior, entailing also high-quality force tracking while in contact. All control details are described in this paper, allowing easy implementation not only for this specific context, but also for other telemanipulation architectures that require remote delicate handling, without jeopardizing force control and orientation control performances.**

***Index Terms*— Contact modeling, force control, medical robots and systems, null-space control, robotic-assisted tele-echography.**

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I. Introduction

THE discovery of X-rays by Wilhelm Röntgen in 1895 was a key milestone for medical science, marking the beginning of a new era in medical diagnosis [1]. For the first time, physicians were able to "look" inside the human body without having to open or inserting an instrument into it. Since then, several other medical imaging techniques have emerged. Medical ultrasound is one of those techniques. It can be seen as a form of "medical sonar," where an ultrasound pulse is sent through the body and an image of traversed structures is built from reflected echoes. Nonetheless, the first medical use of ultrasounds was not as a diagnostic tool. Using the heating and disruptive effects of high-intensity ultrasounds on animal tissue, ultrasounds were first employed as a neurosurgical tool and as a physical therapeutic tool [2]. The first attempt to use ultrasounds as a diagnostic tool was performed by Karl Dussik in 1942, when he attempted to locate brain tumors and cerebral ventricles by measuring the attenuation of an ultrasound beam through the skull [3]. With the maturity of ultrasound, as a physical therapeutic tool and the proliferation of ultrasound devices after World War II, several researchers started to investigate its use as a diagnostic tool. In earlier 1950s, John Wild and John Reid developed a linear hand-held B-mode ultrasound scanner and, by sweeping the scanner from side to side through breast lumps were able to visualize breast tumors [2]. This achievement was a key milestone, demonstrating the potential of ultrasounds for real-time imaging of the interior of the human body in a safe, noninvasive, and relatively inexpensive way. Ever since, ultrasound imaging has never stopped evolving, becoming a widespread and crucial diagnostic tool for a large number of medical specialties. However, it is a procedure that

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requires skilled and experienced physicians, which might not always be available, especially in emergency scenarios or in geographically remote areas. Furthermore, some procedures require the physician to perform, for extended periods of time, forceful and repetitive movements in uncomfortable body postures, leading to a substantial prevalence of work-related musculoskeletal disorders among sonographers [4].

A robotic-assisted tele-echography system pops up as a solution to these problems. By controlling an ultrasound probe through a haptic device, the physician is able to perform the examination in a similar way to the standard procedure. But, to be adopted by the medical community, the probe motion must accurately replicate physician's movements, either in free space or under contact constraints, avoiding force peaks and jerky behaviors when moving from free space to contact and vice versa. Furthermore, the dynamic behavior has to be adapted according to the medical procedure, imposing interaction dynamics constraints.

The control strategies presented in this paper are a natural evolution of authors' previous work. In [5], a simple task-space PD controller was used. Even though good position control was achieved, the orientation control referred to the end-effector had poor results. This problem was addressed in [6] where a joint space velocity controller driven by task-space posture errors was proposed, allowing to control each joint individually. Both approaches established the interaction dynamics based on the equivalence between PD and impedance control, where better motion tracking performance means higher control gains which, in turn, lead to stiffer behavior in contact. In [7], an admittance control was used, outputting a compliant position reference for an inner motion controller. The orientation was controlled without the admittance loop, since, at orientation level, a too compliant behavior may be counterproductive due to tracking errors. However, the compliant design of [7] does not take into account the environment stiffness, providing no adaptation in contact. In addition, couplings between position and orientation exist due to robot modeling errors, affecting control performance. To tackle these issues, we propose a hierarchical control architecture, which relies on the following.

1) Explicit Cartesian force control driven by position errors as the high priority task. The force control establishes a suitable dynamic behavior between probe and patient, limiting the range of applied forces.
2) Probe orientation control in the null space, reducing coupling effects. The orientation control is driven by task-space orientation errors converted into joint velocity references, allowing to control each joint individually, boosting orientation tracking performance.

Both force and orientation are controlled through active observers (AOBs) [8]. Since force control follows a model-reference approach, the control model must be in accordance with the environment (free space, soft, and stiff contact), being adapted whenever the environment changes, to boost control performance. The wrist force sensor where the probe is attached measures not only interaction forces but also inertial forces which are particularly relevant in free space motion. In addition, the inertial properties perceived at the end-effector change with the manipulator posture, changing the contact dynamics, which has impact in the way the controller perceives the environment. To cope with all these issues, we propose the following.

1) Motion dynamics driven by a force sensor and a 3-D time-of-flight camera. When moving in free space toward contact, the environment model is anticipated by 3-D camera data, allowing precontact adaptation. In addition, free space inertial forces are also eliminated from the control loop, improving free space control performance.
2) Control adaptation in contact based on online environment stiffness estimation using force data and robot inertial properties referred to the end-effector. A strong correlation between the perceived stiffness by the controller and the robot effective mass exists, being this correlation used in the stiffness estimation algorithm, improving force tracking performance.

The hierarchical control architecture and the motion dynamics driven by the 3-D camera have already been discussed in [9] and [10], respectively. This paper expands these works by presenting a new stiffness estimation algorithm and joint control linearization in the null space. Furthermore, this paper also assesses and validates the control architecture and the new stiffness estimation strategy in both clinical and nonclinical scenarios. The major novelty is the stiffness estimation algorithm which relies on force relations [8], and also on the manipulator inertial properties referred to the end-effector. The robot effective mass in the estimation algorithm allows anticipating the stiffness perceived by the controller, improving force control performance in dynamic interactions, specially with stiff surfaces. Clearly, the elastic properties of the object do not change during probe interaction; however, the controller perception of the object does change due to a strong correlation between the effective mass and the perceived stiffness. Since contact is modeled as a linear spring, its estimation reflects not only the environment stiffness but also the effect of unmodeled disturbances such as environment nonlinearities, dynamic modeling errors, and nonuniform contact area. In this way, the perceived stiffness can be seen, not as a well defined physical property, but as an optimization to improve force control performance. The environment is modeled as a simple linear spring since in highly unstructured environments, as is an ultrasound procedure, where free space motion interchanges with soft and stiff contact scenarios, simpler but less accurate contact models might be preferable. Merging contact dynamics into a single parameter simplifies estimation algorithms, allowing for real-time contact parameter adaptation. Experimental results validate this approach, with improved force control performance. In this way, the choice of a linear spring contact model is adequate.

The remainder of this paper is organized as follows. Section II presents the literature review. Section III presents manipulator dynamics and strategies to achieve decoupled force and orientation control. The force control architecture and the adaptation strategy are discussed in Section IV. In Section V, the motion control approach is analyzed. Online stiffness estimation is introduced in Section VI, and in

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Section VII experimental results are presented. Section VIII discusses paper merits and future research directions, and Section IX concludes this paper.

II. Related Work

A. Control Architectures for Robotic-Assisted Ultrasound Systems

One of the first approaches to introduce a robot in the echographic procedure was presented by Pierrot *et al.* [11], where a force controlled manipulator was used to autonomously scan an artery for offline 3-D reconstruction. The control architecture features an inner joint position loop, where joint references are computed from its task-space counterpart through inverse kinematics. Interaction forces are controlled by an outer force loop, whose output acts on the reference trajectory to cancel the force error. Zhu *et al.* [12] devised a fully counter-balanced six-degrees of freedom (DoF) robot to be used as the probe holder. The control architecture presents a joint velocity control loop, with the joint velocity reference computed from its task-space counterpart through differential inverse kinematics. By suitable design of a Cartesian velocity reference, in plane ultrasound visual servoing [13], force, and position control are all possible to be accomplished for each dimension individually. Switching between control modes is made offline. Koizumi *et al.* [14] developed a robotic-assisted tele-ecography system. The slave manipulator is impedance controlled at position level, and a continuous path controller is used for orientation [15]. The continuous path control works by sampling and sending together, at a frequency f, n orientation samples from the master to the slave robot, where an orientation trajectory is generated by a cubic spline curve. The authors claim better orientation tracking performance when compared with the traditional point-to-point method, common in master–slave systems. However, it comes at a cost of an extra delay introduced by the orientation sampling. In [16], the probe motion is controlled by a six DoF parallel robot. The robot is position controlled by a PID at joint level. A solution based on a commercially available robot is proposed in [17]. They propose to control the slave manipulator implementing a compliance control loop on top of an internal joint velocity controller. Mylonas *et al.* [18] presented a different solution for remote robotic-assisted ultrasound diagnosis. Focusing in detecting intracavitary hemorrhage, they propose to perform an autonomous ultrasound scan via a learning-based control approach from expert demonstration. The manipulator is controlled by a Cartesian impedance control. Nadeau and Krupa [19] presented an ultrasound intensity-based approach to control a robotic-assisted ultrasound system. The control architecture features a hybrid vision/force controller, with the motion normal to the contact surface controlled at force level while the remaining DoFs are controlled by ultrasound visual servoing. In [20], the proposed technique is applied for instrument and tissue tracking in beating heart surgery guided by 3-D ultrasound.

B. Hierarchical Priority Control

To effectively perform in a human-like environment (highly unstructured), where multiple and, sometimes, simultaneous tasks must be carried out, robots started to be designed with multiple redundant DoFs. Redundancy allows optimization based on the Jacobian null space [21]. Under this general framework, redundancy can either be resolved at velocity [22], acceleration [23], or force level [24]. Hierarchical priority control is a well-established framework to perform multiple tasks simultaneously. By specifying multiple operational points on the robot structure, it is possible to assign different priorities to different tasks both at kinematic [25]–[27] and dynamic levels [28]–[31]. Compliant task-space control presents a suitable approach to create an adequate motion of the end-effector in environments shared with humans. In [32], a study of different task-space controllers for redundant manipulators has been presented. Null-space compliant control has been investigated in [30] while multi priority impedance control has been presented in [33].

C. Payload Inertial Forces

When coupling a load in a wrist force sensor, the measured forces reflect not only contact forces but also inertial forces, which can become dominant in dynamic scenarios, deteriorating force control performance. In [34], a sensor fusing approach (force, acceleration, and position measurements) based on extended Kalman filter is used to estimate noncontact forces, being this approach further extended in [35], where all inertial payload parameters are estimated online. On the other hand, Aghili [36] proposes an impedance control scheme for manipulators carrying heavy payloads that overcomes noncontact forces without performing inertial forces estimation.

D. Environment Model Estimation

Environment model estimation plays an important role in robotic tasks, improving model-reference force control performance, enabling perception of the environment, and allowing computer simulation of contact scenarios. Usually, the environment is modeled as a linear spring (elastic model) or as a linear spring-damper model [Kelvin–Voight (KV) model]. In [37], a recursive least-squares (RLS) technique adapts the parameters of an impedance controller to environment changes. Adaptive control techniques are used for stiffness estimation in [38] and for stiffness and damping estimation in [39]. Erickson *et al.* [40] perform a comparative study between the methods presented in [37]–[39] and an approach proposed by them, in which the KV model parameters are estimated by an offline signal processing method. In [41], an RLS technique estimates the contact stiffness taking into account force measurements delays, while in [42] an exponentially weighted RLS is used for stiffness estimation in a force controlled industrial robot. Environment linear models are simpler to estimate, however, they present physical inconsistencies. To tackle these inconsistencies, environment estimation based on nonlinear models has also been addressed. Diollati *et al.* [43] show that the Hunt–Crossley (HC) model has a better physical consistency then the KV model, proposing a double-stage RLS approach for online contact model estimation. In [44], HC parameter identification is improved by a single-stage method. Verscheure *et al.* [45] propose to

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estimate the contact parameters of an environment modeled by a Hertzian spring using a total least-squares estimator. Previous approaches require the availability of force, position, and, sometimes, velocity measurements, together with a precise knowledge of contact position. For most practical robotic applications, where the environment is unstructured and free space motion interchanges with soft and stiff contact scenarios, such data are almost unpractical. To address this issue, few approaches in the literature (see [8], [46], and [47]), propose to estimate the environment stiffness resorting only to force data.

III. Manipulator Dynamic Model

Given a set of n generalized coordinates $\boldsymbol{q} \in \mathbb{R}^n$ and a set of generalized torques $\boldsymbol{\tau} \in \mathbb{R}^n$, the manipulator dynamic model is given by

$$\boldsymbol{B}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{1}$$

where $\boldsymbol{B}(\boldsymbol{q}) \in \mathbb{R}^{n\times n}$ is the manipulator inertia matrix, $\boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^n$ is the vector of Coriolis and centripetal terms, and $\boldsymbol{g}(\boldsymbol{q}) \in \mathbb{R}^n$ is the gravity term.

A. Nonlinear Feedback Linearization

The manipulator formulated in (1) is a nonlinear system. If the dynamic model is known [48], a linear and decoupled system can be achieved through nonlinear feedback linearization. Let the total torques acting on the manipulator be given by

$$\boldsymbol{\tau} = \boldsymbol{\tau}_c + \boldsymbol{\tau}_f + \boldsymbol{\tau}_m \tag{2}$$

where $\boldsymbol{\tau}_c \in \mathbb{R}^n$, $\boldsymbol{\tau}_f \in \mathbb{R}^n$, and $\boldsymbol{\tau}_m \in \mathbb{R}^n$ are actuator, friction, and interaction torques, respectively. Neglecting joint frictions ($\boldsymbol{\tau}_f \approx 0$) and defining $\boldsymbol{\tau}_m$ as the torques due to interaction

$$\boldsymbol{\tau}_m = \boldsymbol{J}^T(\boldsymbol{q})\boldsymbol{f}_{\text{fs}} \tag{3}$$

where $\boldsymbol{f}_{\text{fs}} \in \mathbb{R}^m$ is a measurement of end-effector forces and $\boldsymbol{J}^T(\boldsymbol{q}) \in \mathbb{R}^{n\times m}$ is the Jacobian transpose, the commanded torque $\boldsymbol{\tau}_c$ can be defined as[1]

$$\boldsymbol{\tau}_c = \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) - \boldsymbol{\tau}_m + \boldsymbol{\tau}'_c. \tag{4}$$

From (4) and (1), the system reduces to

$$\boldsymbol{B}(\boldsymbol{q})\ddot{\boldsymbol{q}} = \boldsymbol{\tau}'_c \tag{5}$$

where $\boldsymbol{\tau}'_c \in \mathbb{R}^n$ is the torque vector computed by the control law. Equation (5) can be represented in the task space [24] as

$$\boldsymbol{\Lambda}_p(\boldsymbol{q})\ddot{\boldsymbol{x}}_p - \boldsymbol{\Lambda}_p(\boldsymbol{q})\dot{\boldsymbol{J}}_p(\boldsymbol{q})\dot{\boldsymbol{q}} = \boldsymbol{f}_p \tag{6}$$

with

$$\dot{\boldsymbol{x}}_p = \boldsymbol{J}_p(\boldsymbol{q})\dot{\boldsymbol{q}} \tag{7}$$

$$\ddot{\boldsymbol{x}}_p = \boldsymbol{J}_p(\boldsymbol{q})\ddot{\boldsymbol{q}} + \dot{\boldsymbol{J}}_p(\boldsymbol{q})\dot{\boldsymbol{q}} \tag{8}$$

$$\boldsymbol{\Lambda}_p(\boldsymbol{q}) = \left(\boldsymbol{J}_p(\boldsymbol{q})\boldsymbol{B}^{-1}(\boldsymbol{q})\boldsymbol{J}_p^T(\boldsymbol{q})\right)^{-1} \tag{9}$$

$$\boldsymbol{\tau}'_c = \boldsymbol{\tau}_p = \boldsymbol{J}_p^T(\boldsymbol{q})\boldsymbol{f}_p. \tag{10}$$

where $\boldsymbol{\Lambda}_p(\boldsymbol{q}) \in \mathbb{R}^{m\times m}$ and $\boldsymbol{J}_p(\boldsymbol{q}) \in \mathbb{R}^{m\times n}$ are the inertia and the Jacobian matrix associated with task $\boldsymbol{x}_p \in \mathbb{R}^m$. $\boldsymbol{\tau}_p \in \mathbb{R}^n$ and $\boldsymbol{f}_p \in \mathbb{R}^m$ are, respectively, torque and force vectors computed by the controller to perform $\boldsymbol{x}_p$. Setting $\boldsymbol{f}_p$ in (6) as

$$\boldsymbol{f}_p = \boldsymbol{\Lambda}_p(\boldsymbol{q})\boldsymbol{f}^* - \boldsymbol{\Lambda}_p(\boldsymbol{q})\dot{\boldsymbol{J}}_p(\boldsymbol{q})\dot{\boldsymbol{q}} \tag{11}$$

a linear and decoupled system arises, representing the dynamics of a unitary mass for each Cartesian dimension

$$\ddot{\boldsymbol{x}}_p = \boldsymbol{f}^* \tag{12}$$

where $\boldsymbol{f}^* \in \mathbb{R}^m$ is the new control variable.

B. Null-Space Control

If $m < n$, the system is redundant when performing $\boldsymbol{x}_p$ and $\boldsymbol{\tau}'_c$ are not unique. The generalized torque/force relationship is given by

$$\boldsymbol{\tau}'_c = \boldsymbol{J}_p^T(\boldsymbol{q})\boldsymbol{f}_p + \boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{\tau}_s. \tag{13}$$

This relationship allows the decomposition of $\boldsymbol{\tau}'_c$ into two dynamically decoupled torque vectors

$$\boldsymbol{\tau}'_c = \boldsymbol{\tau}_p + \boldsymbol{\tau}_n \tag{14}$$

being $\boldsymbol{\tau}_p$ the torque performing the primary task [see (10)] and

$$\boldsymbol{\tau}_n = \boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{\tau}_s \tag{15}$$

the torque of a secondary task that does not interfere with $\boldsymbol{\tau}_p$ and is performed in the null space of $\boldsymbol{J}_p^T(\boldsymbol{q})$

$$\boldsymbol{N}_p^T(\boldsymbol{q}) = [\boldsymbol{I} - \bar{\boldsymbol{J}}_p(\boldsymbol{q})\boldsymbol{J}_p(\boldsymbol{q})]^T = \left[\boldsymbol{I} - \boldsymbol{J}_p^T(\boldsymbol{q})\bar{\boldsymbol{J}}_p^T(\boldsymbol{q})\right] \tag{16}$$

where $\boldsymbol{N}_p^T(\boldsymbol{q}) \in \mathbb{R}^{n\times n}$ is the null-space projector of $\boldsymbol{x}_p$, $\boldsymbol{I} \in \mathbb{R}^{n\times n}$ the identity matrix and $\bar{\boldsymbol{J}}_p(\boldsymbol{q}) \in \mathbb{R}^{m\times n}$ is the dynamically consistent generalized inverse of $\boldsymbol{J}_p(\boldsymbol{q})$ [24], given by

$$\bar{\boldsymbol{J}}_p(\boldsymbol{q}) = \boldsymbol{B}^{-1}(\boldsymbol{q})\boldsymbol{J}_p^T(\boldsymbol{q})\boldsymbol{\Lambda}_p(\boldsymbol{q}). \tag{17}$$

Let $\boldsymbol{x}_s \in \mathbb{R}^l$ be a secondary task

$$\dot{\boldsymbol{x}}_s = \boldsymbol{J}_s(\boldsymbol{q})\dot{\boldsymbol{q}} \tag{18}$$

where $\boldsymbol{J}_s(\boldsymbol{q}) \in \mathbb{R}^{l\times n}$ is the respective Jacobian matrix. The torque to perform the secondary task $\boldsymbol{\tau}_s \in \mathbb{R}^n$ can be defined similarly to $\boldsymbol{\tau}_p$ as

$$\boldsymbol{\tau}_s = \boldsymbol{J}_s^T(\boldsymbol{q})\boldsymbol{f}_s \tag{19}$$

where $\boldsymbol{f}_s \in \mathbb{R}^l$ is the task-space counterpart. Substituting (19) in (13)

$$\begin{aligned}\boldsymbol{\tau}'_c &= \boldsymbol{J}_p^T(\boldsymbol{q})\boldsymbol{f}_p + \boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{J}_s^T(\boldsymbol{q})\boldsymbol{f}_s \\ &= \boldsymbol{J}_p^T(\boldsymbol{q})\boldsymbol{f}_p + \boldsymbol{J}_{s|p}^T(\boldsymbol{q})\boldsymbol{f}_s\end{aligned} \tag{20}$$

where

$$\boldsymbol{J}_{s|p}^T(\boldsymbol{q}) = (\boldsymbol{J}_s(\boldsymbol{q})\boldsymbol{N}_p(\boldsymbol{q}))^T \tag{21}$$

appears as a new Jacobian matrix [28], combining the primary task null-space operator with the secondary task Jacobian. Its range space is the instantaneous space of the secondary task motion that is consistent with the primary task.

[1]It is assumed throughout the paper that robot kinematic and dynamic parameters are estimated with negligible errors.

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If $m+l < n$, the system is still redundant when performing $\boldsymbol{x}_p$ and $\boldsymbol{x}_s$. The generalized $\boldsymbol{\tau}_c'$ is then given by

$$\begin{aligned}\boldsymbol{\tau}_c' &= \boldsymbol{J}_p^T(\boldsymbol{q})\boldsymbol{f}_p + \boldsymbol{J}_{s|p}^T(\boldsymbol{q})\boldsymbol{f}_s + \boldsymbol{N}_N^T(\boldsymbol{q})\boldsymbol{\tau}_0 \\ &= \boldsymbol{\tau}_p + \boldsymbol{\tau}_{s|p} + \boldsymbol{N}_N^T(\boldsymbol{q})\boldsymbol{\tau}_0 \end{aligned} \tag{22}$$

where $\boldsymbol{\tau}_{s|p}$ is the torque to perform the secondary task constrained by the primary task, $\boldsymbol{\tau}_0 \in \mathbb{R}^n$ is a performance optimization torque, and $\boldsymbol{N}_N(\boldsymbol{q}) \in \mathbb{R}^{n\times n}$ is the projection matrix for the null space of all prioritized tasks, given by

$$\begin{aligned}\boldsymbol{N}_N(\boldsymbol{q}) &= \boldsymbol{N}_p(\boldsymbol{q})\boldsymbol{N}_{s|p}(\boldsymbol{q}) \\ &= (\boldsymbol{I} - \bar{\boldsymbol{J}}_p(\boldsymbol{q})\boldsymbol{J}_p(\boldsymbol{q}))(\boldsymbol{I} - \bar{\boldsymbol{J}}_{s|p}(\boldsymbol{q})\boldsymbol{J}_{s|p}(\boldsymbol{q})) \\ &= \boldsymbol{I} - \bar{\boldsymbol{J}}_p(\boldsymbol{q})\boldsymbol{J}_p(\boldsymbol{q}) - \bar{\boldsymbol{J}}_{s|p}(\boldsymbol{q})\boldsymbol{J}_{s|p}(\boldsymbol{q}) \end{aligned} \tag{23}$$

where the property $\boldsymbol{J}_p\boldsymbol{N}_p(\boldsymbol{q}) = \boldsymbol{0}$, with $\boldsymbol{0} \in \mathbb{R}^{n\times n}$ as the zero matrix, has been used.

C. Explicit Force Control With Null-Space Orientation Control

Let $\boldsymbol{\tau}_p$ be the torque computed by the force controller and $\boldsymbol{\tau}_s$ be the torque computed by the joint space orientation controller, where $\boldsymbol{J}_p(\boldsymbol{q})$ and $\boldsymbol{J}_s(\boldsymbol{q})$ are the position and the orientation Jacobian, respectively. Let also $\boldsymbol{f}^* \in \mathbb{R}^3$ be the control variable for the primary task and $\boldsymbol{\alpha} \in \mathbb{R}^n$ be the control variable for the secondary task. From (5) and (22), the manipulator dynamics with Coriolis, gravity, and external force precompensation in the joint space [see (4)] is given by

$$\boldsymbol{B}(\boldsymbol{q})\ddot{\boldsymbol{q}} = \boldsymbol{\tau}_p + \boldsymbol{\tau}_{s|p} + \boldsymbol{N}_N^T(\boldsymbol{q})\boldsymbol{\tau}_0. \tag{24}$$

Left multiplying both sides of (24) by $\boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})$ and setting $\boldsymbol{\tau}_{s|p}$ as

$$\boldsymbol{\tau}_{s|p} = \boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{\tau}_s = \boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{B}(\boldsymbol{q})\boldsymbol{\alpha} \tag{25}$$

leads to

$$\begin{aligned}&\boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})\boldsymbol{B}(\boldsymbol{q})\ddot{\boldsymbol{q}} \\ &\quad= \boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})\boldsymbol{\tau}_p + \boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})\boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{B}(\boldsymbol{q})\boldsymbol{\alpha} \\ &\quad+ \underbrace{\boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})\boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{N}_{s|p}^T(\boldsymbol{q})}_{=\boldsymbol{0}}\boldsymbol{\tau}_0 \end{aligned} \tag{26}$$

where $\bar{\boldsymbol{J}}_s(\boldsymbol{q}) \in \mathbb{R}^{3\times n}$ is the dynamically consistent generalized inverse of $\boldsymbol{J}_s(\boldsymbol{q}) \in \mathbb{R}^{3\times n}$, computed similarly to (17). $\boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})\boldsymbol{B}(\boldsymbol{q})\ddot{\boldsymbol{q}}$ are the control torques acting on the secondary task space, composed by the orientation control torques [second term of (26)] and the coupling torques from the primary task (first term). The third term vanish due to the presence of $\boldsymbol{J}_{s|p}(\boldsymbol{q})\boldsymbol{N}_{s|p}(\boldsymbol{q}) = \boldsymbol{0}$

$$\begin{aligned}&\boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})\boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{N}_{s|p}^T(\boldsymbol{q})\boldsymbol{\tau}_0 \\ &\qquad= \boldsymbol{B}(\boldsymbol{q})\bar{\boldsymbol{J}}_s(\boldsymbol{q})\underbrace{\boldsymbol{J}_s(\boldsymbol{q})\boldsymbol{N}_p}_{=\boldsymbol{J}_{s|p}(\boldsymbol{q})}(\boldsymbol{q})\boldsymbol{N}_{s|p}(\boldsymbol{q})\ddot{\boldsymbol{q}}_0 \end{aligned} \tag{27}$$

where the commutation properties have been used

$$\boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{B}(\boldsymbol{q}) = \boldsymbol{B}(\boldsymbol{q})\boldsymbol{N}_p(\boldsymbol{q}) \tag{28}$$

$$\boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T\boldsymbol{B}(\boldsymbol{q}) = \boldsymbol{B}(\boldsymbol{q})\bar{\boldsymbol{J}}_s(\boldsymbol{q})\boldsymbol{J}_s(\boldsymbol{q}). \tag{29}$$

Projecting (26) through $\boldsymbol{N}_p^T(\boldsymbol{q})$, the torques acting on the secondary task space without affecting the primary task execution are given by

$$\begin{aligned}&\boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})\boldsymbol{B}(\boldsymbol{q})\ddot{\boldsymbol{q}} \\ &\quad= \boldsymbol{N}_p^T(\boldsymbol{q}))\boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})\boldsymbol{\tau}_p \\ &\qquad+ \boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})\boldsymbol{N}_p^T(\boldsymbol{q})\boldsymbol{B}(\boldsymbol{q})\boldsymbol{\alpha}. \end{aligned} \tag{30}$$

Left multiplying both sides by $\boldsymbol{B}^{-1}(\boldsymbol{q})$ and using the commutation properties

$$\boldsymbol{B}^{-1}(\boldsymbol{q})\boldsymbol{N}_p^T(\boldsymbol{q}) = \boldsymbol{N}_p(\boldsymbol{q})\boldsymbol{B}^{-1}(\boldsymbol{q}) \tag{31}$$

$$\boldsymbol{B}^{-1}(\boldsymbol{q})\boldsymbol{J}_s^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T = \bar{\boldsymbol{J}}_s(\boldsymbol{q})\boldsymbol{J}_s(\boldsymbol{q})\boldsymbol{B}^{-1}(\boldsymbol{q}) \tag{32}$$

the corresponding secondary task joint acceleration is given by

$$\begin{aligned}\boldsymbol{N}_p(\boldsymbol{q})\bar{\boldsymbol{J}}_s(\boldsymbol{q})\boldsymbol{J}_s(\boldsymbol{q})\ddot{\boldsymbol{q}} &= \boldsymbol{N}_p(\boldsymbol{q})\bar{\boldsymbol{J}}_s(\boldsymbol{q})\boldsymbol{J}_s(\boldsymbol{q})\boldsymbol{B}^{-1}(\boldsymbol{q})\boldsymbol{\tau}_p \\ &\quad+ \boldsymbol{N}_p(\boldsymbol{q})\bar{\boldsymbol{J}}_s(\boldsymbol{q})\boldsymbol{J}_{s|p}(\boldsymbol{q})\boldsymbol{\alpha}. \end{aligned} \tag{33}$$

Setting the control variable $\boldsymbol{\alpha}$ as

$$\boldsymbol{\alpha} = \bar{\boldsymbol{J}}_{s|p}(\boldsymbol{q})\boldsymbol{J}_s(\boldsymbol{q})(\boldsymbol{\nu} - \boldsymbol{B}^{-1}(\boldsymbol{q})\boldsymbol{\tau}_p) \tag{34}$$

and replacing (34) in (33), the secondary task closed-loop dynamics becomes

$$\boldsymbol{N}_p(\boldsymbol{q})\bar{\boldsymbol{J}}_s(\boldsymbol{q})\boldsymbol{J}_s(\boldsymbol{q})(\ddot{\boldsymbol{q}} - \boldsymbol{\nu}) = \boldsymbol{0} \tag{35}$$

where $\boldsymbol{\nu} \in \mathcal{R}^n$ arises as a new control variable. A linear and decoupled joint acceleration in the primary task null space shows up, performing the secondary task without inducing any motion in the primary task space.

Plugging (34) in (25) and replacing $\bar{\boldsymbol{J}}_{s|p}(\boldsymbol{q})$ by

$$\bar{\boldsymbol{J}}_{s|p}(\boldsymbol{q}) = \boldsymbol{B}^{-1}(\boldsymbol{q})\boldsymbol{J}_{s|p}^T(\boldsymbol{q})\boldsymbol{\Lambda}_{s|p}(\boldsymbol{q}). \tag{36}$$

$\boldsymbol{\tau}_{s|p}$ becomes

$$\boldsymbol{\tau}_{s|p} = \boldsymbol{J}_{s|p}^T(\boldsymbol{q})\boldsymbol{\Lambda}_{s|p}(\boldsymbol{q})\boldsymbol{J}_s(\boldsymbol{q})(\boldsymbol{\nu} - \boldsymbol{B}^{-1}(\boldsymbol{q})\boldsymbol{\tau}_p) \tag{37}$$

where the null-space idempotence property $\boldsymbol{N}_p^T(\boldsymbol{q}) = (\boldsymbol{N}_p^T(\boldsymbol{q}))^2$ has been used and $\boldsymbol{\Lambda}_{s|p}(\boldsymbol{q})$ is a prioritized inertia matrix [28] given by

$$\boldsymbol{\Lambda}_{s|p}(\boldsymbol{q}) = \left(\boldsymbol{J}_s(\boldsymbol{q})\boldsymbol{B}^{-1}(\boldsymbol{q})\boldsymbol{J}_{s|p}^T(\boldsymbol{q})\right)^{-1}. \tag{38}$$

Substituting (37) and (11) in (22), the control torque to perform both tasks is given by

$$\begin{aligned}\boldsymbol{\tau}_c' &= \boldsymbol{J}_p^T(\boldsymbol{q})[\boldsymbol{\Lambda}_p(\boldsymbol{q})\boldsymbol{f}^* - \boldsymbol{\Lambda}_p(\boldsymbol{q})\dot{\boldsymbol{J}}_p(\boldsymbol{q})\dot{\boldsymbol{q}}] \\ &\quad+ \boldsymbol{J}_{s|p}^T(\boldsymbol{q})\boldsymbol{\Lambda}_{s|p}(\boldsymbol{q})\boldsymbol{J}_s(\boldsymbol{q})[\boldsymbol{\nu} - \boldsymbol{B}^{-1}(\boldsymbol{q})\boldsymbol{\tau}_p] + \boldsymbol{N}_N^T\boldsymbol{\tau}_0. \end{aligned} \tag{39}$$

In the above formulation, algorithmic singularities arise when the primary and the secondary task are in conflict, causing $\boldsymbol{J}_{s|p}^T(\boldsymbol{q})$ to drop rank [49]. To avoid issues with such singularities, $\boldsymbol{\alpha}$ is computed by

$$\boldsymbol{\alpha} = \bar{\boldsymbol{J}}_s(\boldsymbol{q})\boldsymbol{J}_s(\boldsymbol{q})(\boldsymbol{\nu} - \boldsymbol{B}^{-1}(\boldsymbol{q})\boldsymbol{\tau}_p) \tag{40}$$

where $\bar{\boldsymbol{J}}_s(\boldsymbol{q})$ has been used instead of $\bar{\boldsymbol{J}}_{s|p}(\boldsymbol{q})$. $\boldsymbol{J}_s(\boldsymbol{q})$ is the orientation Jacobian computed without any constraints from the force controller, with all DoFs available. Due to the manipulator structure, $\boldsymbol{J}_s(\boldsymbol{q})$ only loses rank at the external boundary of its reachable workspace, having no ill-condition

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issues near internal singular configurations, when both tasks are in conflict. Following this approach and using (28) and (39) becomes

$$\boldsymbol{\tau}_c' = \boldsymbol{J}_p^T(\boldsymbol{q})[\boldsymbol{\Lambda}_p(\boldsymbol{q})\boldsymbol{f}^* - \boldsymbol{\Lambda}_p(\boldsymbol{q})\dot{\boldsymbol{J}}_p(\boldsymbol{q})\dot{\boldsymbol{q}}] + \boldsymbol{J}_{s|p}^T(\boldsymbol{q})\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})[\boldsymbol{B}(\boldsymbol{q})\boldsymbol{v} - \boldsymbol{\tau}_p] + \boldsymbol{N}_N^T\boldsymbol{\tau}_0. \quad (41)$$

The closed-loop system in (35) is no longer achievable. As a consequence, tracking errors might arise in the orientation control, since the forces given by $\bar{\boldsymbol{J}}_s^T(\boldsymbol{q})[\boldsymbol{B}(\boldsymbol{q})\boldsymbol{v} - \boldsymbol{\tau}_p]$ might have components along the primary task space, which are canceled out by $\boldsymbol{N}_p^T(\boldsymbol{q})$. However, as it is shown in Section VII-D, this approach together with the orientation control strategy provides good orientation tracking results in the presence or absence of contact constraints.

D. Posture Optimization

The manipulator posture plays a key role in performance optimization. In our setup, performance is optimized by keeping redundant joints close to the middle of their range value through gradient minimization, projecting $\boldsymbol{\tau}_0$ in the null space of all prioritized tasks [see (23)]

$$\boldsymbol{\tau}_0 = \boldsymbol{B}(\boldsymbol{q})\boldsymbol{K}_N\frac{\partial v(\boldsymbol{q})}{\partial \boldsymbol{q}} \quad (42)$$

where $\boldsymbol{K}_N \in \mathcal{R}^{n\times n}$ is a diagonal gain matrix and $v(\boldsymbol{q})$ is an objective function given by

$$v(\boldsymbol{q}) = -\frac{1}{2}\sum_{i=1}^{n}\left(\frac{q_i - \bar{q}_i}{q_{i_M} - q_{i_m}}\right)^2 \quad (43)$$

where $\bar{q}_i$, q_{i_M} and q_{i_m} are, respectively, the ith joint middle, maximum and minimum range value.

Projecting $\boldsymbol{\tau}_0$ through (23) in (41), the total computed torque is then given by replacing (41) in (4). Fig. 1 shows the proposed control architecture scheme, where force and orientation control are the primary and secondary tasks, respectively, and posture optimization is performed in the null space of all prioritized tasks.

IV. Force Control

This section discusses the force control architecture. Each Cartesian force dimension is controlled through an AOB [8]. $\boldsymbol{f}_d \in \mathbb{R}^3$ is the force reference, computed from the position error through virtual coupling as

$$\boldsymbol{f}_d = \boldsymbol{K}_{\text{virt}}(\boldsymbol{p}_d - \boldsymbol{p}_r) \quad (44)$$

where $\boldsymbol{p}_d \in \mathbb{R}^3$ and $\boldsymbol{p}_r \in \mathbb{R}^3$ are, respectively, desired and end-effector Cartesian positions and $\boldsymbol{K}_{\text{virt}} \in \mathbb{R}^{3\times 3}$ is the virtual coupling matrix (see Section VI-C). In the sequel, the control analysis is done for each Cartesian force dimension individually.

A. System Plant

Equation (12) describes the dynamics of a unitary mass for each Cartesian dimension. Modeling the environment as single contact point linear spring with nominal stiffness $K_{s,n}$, and assuming a small time system delay T_d, the plant of a force controlled robot can be approximated by

$$G_f(s) \approx \frac{K_{s,n}e^{-sT_d}}{s\left(s + K_2^t\right)} \quad (45)$$

where K_2^t is the system damping, which includes the feedback gain K_2^v [50] and the robot structural damping K_2^s

$$K_2^t = K_2^v + K_2^s. \quad (46)$$

The equivalent temporal representation is given by

$$\ddot{y}(t) + K_2^t\dot{y}(t) = K_{s,n}u(t - T_d) \quad (47)$$

where $u(t)$ is an acceleration (plant input) and $y(t)$ is the force exerted by the robot and measured at the end-effector (plant output). Defining $x_{r1}(t) = y(t)$ and $x_{r2}(t) = \dot{y}(t)$, (47) can be written in a state-space representation as

$$\begin{cases}\dot{\boldsymbol{x}}_r(t) = \boldsymbol{A}\boldsymbol{x}_r(t) + \boldsymbol{B}\boldsymbol{u}(t - T_d)\\ y(t) = \boldsymbol{C}\boldsymbol{x}_r(t)\end{cases} \quad (48)$$

with

$$\boldsymbol{A} = \begin{bmatrix}0 & 1\\ 0 & -K_2^t\end{bmatrix} \quad \boldsymbol{B} = \begin{bmatrix}0\\ K_{s,n}\end{bmatrix} \text{ and } \boldsymbol{C} = \begin{bmatrix}1 & 0\end{bmatrix}.$$

Discretizing (48) with sampling time h, the equivalent discrete time system becomes[2]

$$\begin{cases}\boldsymbol{x}_{r,k} = \boldsymbol{\Phi}_r\boldsymbol{x}_{r,k-1} + \boldsymbol{\Gamma}_r u_{k-1}\\ y_k = \boldsymbol{C}_r\boldsymbol{x}_{r,k}\end{cases} \quad (49)$$

with

$$\boldsymbol{\Phi}_r = \begin{bmatrix}\boldsymbol{\Phi}_1 & \boldsymbol{\Gamma}_1 & \boldsymbol{\Gamma}_0 & \dots & 0\\ 0 & 0 & 1 & \dots & 0\\ \vdots & \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & 0 & \dots & 1\\ 0 & 0 & 0 & \dots & 0\end{bmatrix}. \quad (50)$$

B. Active Observer Design

An AOB is a state-space observer that reformulates the Kalman filter framework to accomplish model-reference adaptive control [8], [51]. Resorting to an explicit pole placement approach to achieve a desired closed-loop response, the AOB uses the stochastic design of the Kalman matrices to reflect the uncertainties of system model and output measurements in the state estimation. An extra state (active state) p_k is employed to estimate and compensate an equivalent disturbance referred to the system input due to unmodeled terms. The system state estimate is given by[3,4]

$$\begin{bmatrix}\hat{\boldsymbol{x}}_{r,k}\\ \hat{p}_k\end{bmatrix} = \begin{bmatrix}\boldsymbol{\Phi}_r - \boldsymbol{\Gamma}_r\boldsymbol{L}_r & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}\hat{\boldsymbol{x}}_{r,k-1}\\ \hat{p}_{k-1}\end{bmatrix} + \begin{bmatrix}\boldsymbol{\Gamma}_r\\ 0\end{bmatrix}\boldsymbol{r}_{k-1} + \boldsymbol{K}_k(y_k - \hat{y}_k) \quad (51)$$

with

$$\hat{y}_k = \boldsymbol{C}_a\left(\begin{bmatrix}\boldsymbol{\Phi}_r - \boldsymbol{\Gamma}_r\boldsymbol{L}_r & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}\hat{\boldsymbol{x}}_{r,k-1}\\ \hat{p}_{k-1}\end{bmatrix} + \begin{bmatrix}\boldsymbol{\Gamma}_r\\ 0\end{bmatrix}\boldsymbol{r}_{k-1}\right) \quad (52)$$

[2]Details on discrete matrices can be seen in [8]

[3]Throughout the paper the symbol ˆ means estimation.

[4]The subscript k indicates the iteration.

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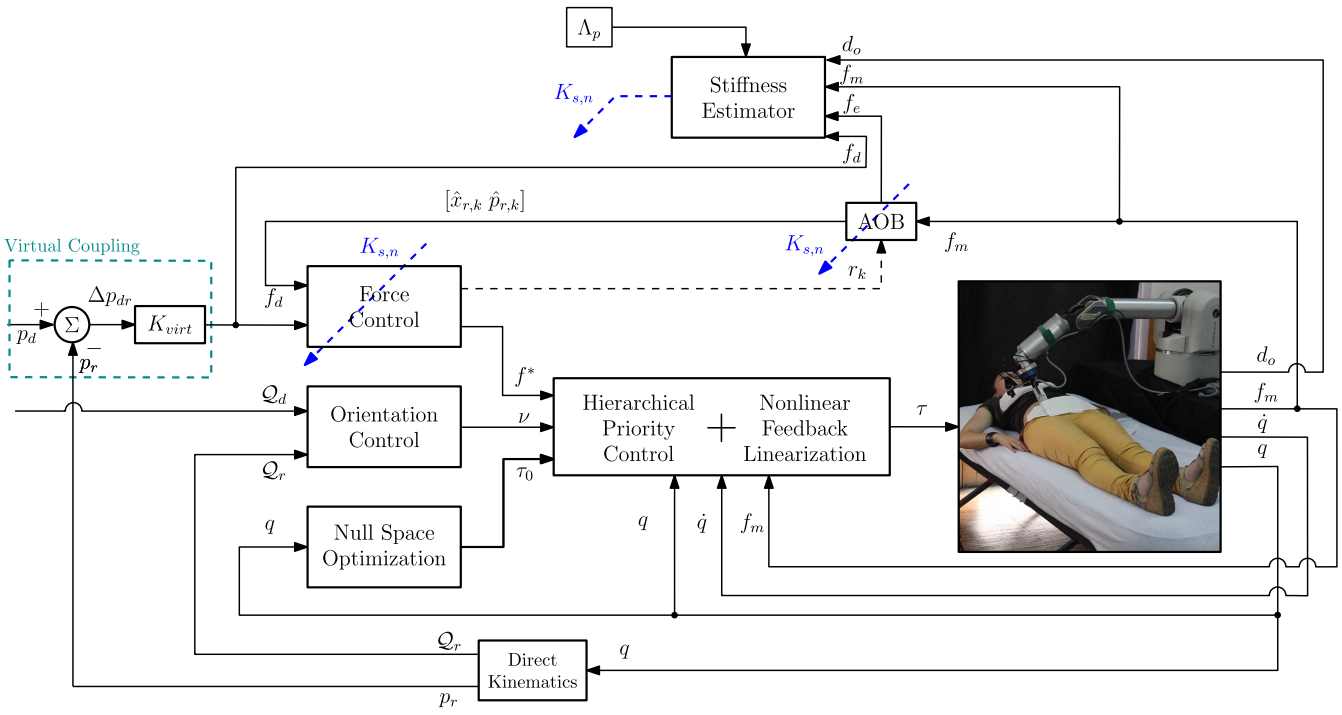


Fig. 1. Hierarchical control architecture for robotic-assisted tele-echography. A two-task hierarchy architecture with posture optimization is proposed, where Cartesian force control is the primary task, orientation control is the secondary task, and posture optimization is performed in the null space of all prioritized tasks. An explicit force control architecture controls the probe positioning and interaction dynamics through a model-reference adaptive control (AOB) in the task space. Online control adaptation is performed based on the environment estimation ($K_{s,n}$). The environment is modeled as a single contact point linear spring, being estimated based on contact distance from a depth camera (d_o), force data (f_d, f_{fs}, and f_e), and the robot inertial properties referred to the end-effector (Λ_p). The blue dashed arrows indicate the control blocks that are updated with $K_{s,n}$ adaptation. Force references are generated from position errors (Δp_{dr}) through virtual coupling (K_{virt}). The probe orientation is controlled by a joint space velocity control driven by task-space orientation errors. Posture is optimized by keeping redundant joints close to the middle of their range value.

where $\boldsymbol{r}_{k-1}$ is the input reference and $\boldsymbol{L}_r$ is the state feedback gain matrix used to set the closed-loop system response. Computed by the well-known Ackermann's formula [52], $\boldsymbol{L}_r$ is set to achieve a critically damped behavior ($\zeta = 1$), with time constant τ_f. The poles due to T_d are mapped in $z = 0$ (discrete-time control representation). The Kalman gain $\boldsymbol{K}_k$ is given by [53]

$$\boldsymbol{K}_k = \boldsymbol{P}_{1k}\boldsymbol{C}_a^T\left[\boldsymbol{C}_a\boldsymbol{P}_{1k}\boldsymbol{C}_a^T + \boldsymbol{R}_k\right]^{-1} \tag{53}$$

with

$$\boldsymbol{P}_{1k} = \boldsymbol{\Phi}_n\boldsymbol{P}_{k-1}\boldsymbol{\Phi}_n^T + \boldsymbol{Q}_k \tag{54}$$

$$\boldsymbol{P}_k = \boldsymbol{P}_{1k} - \boldsymbol{K}_k\boldsymbol{C}_a\boldsymbol{P}_{1k} \tag{55}$$

where $\boldsymbol{\Phi}_n$ is the open-loop augmented state transition matrix and $\boldsymbol{R}_k = E\{\boldsymbol{\kappa}_k\boldsymbol{\kappa}_k^T\}$ and $\boldsymbol{Q}_k = E\{\boldsymbol{\xi}_k\boldsymbol{\xi}_k^T\}$ are, respectively, measurement and system noise matrix

$$\boldsymbol{\Phi}_n = \begin{bmatrix}\boldsymbol{\Phi}_r & \boldsymbol{\Gamma}_r \\ \boldsymbol{0} & 1\end{bmatrix}, \quad \boldsymbol{Q}_k = \begin{bmatrix}\boldsymbol{Q}_{x_{r,k}} & \boldsymbol{0} \\ \boldsymbol{0} & Q_{p_k}\end{bmatrix}. \tag{56}$$

$\boldsymbol{\xi}_k$ and $\boldsymbol{\kappa}_k$ are the stochastic inputs due to model and measure uncertainties, respectively. $\boldsymbol{P}_k$ is the mean-square error matrix and its initial value should not be lower than the initial value of $\boldsymbol{Q}_k$.[5] The force control variable f^* is given by

$$f^* = L_1 f_d - [\boldsymbol{L}_r\ 1]\begin{bmatrix}\hat{\boldsymbol{x}}_{r,k} \\ \hat{p}_k\end{bmatrix}. \tag{57}$$

[5]See [8] for more details on AOB design, stability, and robustness.

where L_1 is the first element of $\boldsymbol{L}_r$. Consistent force responses require that the AOB contact model adapts when the environment changes. If the environment stiffness changes from K_s to $K_s + \Delta K_s$, only Γ_1 needs to be updated in $\boldsymbol{\Phi}_r$ [see (50)] [54]

$$\Gamma_1^{K_s+\Delta K_s} = \Gamma_1^{K_s} + \Delta\Gamma_1 \tag{58}$$

$$\Delta\Gamma_1 = \frac{\Delta K_s}{K_s}\Gamma_1^{K_s}. \tag{59}$$

The system feedback gain also needs to be updated

$$\boldsymbol{L}_r = \begin{bmatrix}\dfrac{L_1}{1+\Delta K_s/K_s} & \dfrac{L_2}{1+\Delta K_s/K_s} & L_3 & \dots & L_n\end{bmatrix} \tag{60}$$

while the gains due to T_d are kept constant. Fig. 2 shows the force control scheme with the AOB in the loop.

C. Inertial Forces

Free space motion leads to nonnegligible inertial forces being captured by a wrist-force sensor when a tool is attached to it. These forces enter in the AOB and in system linearization [see (3)], deteriorating control performance. Using the depth information from a 3-D time-of-flight camera mounted at end-effector in an eye-in-hand configuration, measured forces can be taken into account only near contact, removing free space inertial forces from the control loop. Let $\boldsymbol{P} \in \mathbb{R}^{W\times H}$ be a depth image, where W and H are image width and height, respectively. ${}^c\boldsymbol{p}(i,j) \in \mathbb{R}^4$ is the 3-D homogeneous

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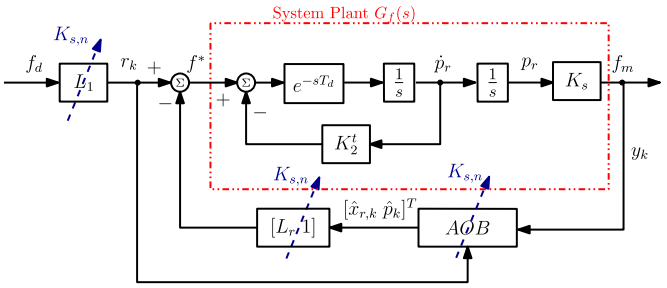


Fig. 2. Force control with an AOB in the loop. Each Cartesian force dimension is controlled by an AOB. The blue dashed arrows show the control variables that are updated due to the stiffness adaptation $K_{s,n}$.

coordinates of the ith, jth camera pixel projected in the camera frame

$$^c\boldsymbol{p}(i,j) = [^c p_x(i,j) \quad ^c p_y(i,j) \quad ^c p_z(i,j) \quad 1]^T.$$

Since the homogeneous transformation $^p\boldsymbol{T}_c \in \mathbb{R}^{4\times4}$ between camera and probe is known, $^c\boldsymbol{p} \in \mathbb{R}^4$ can be represented in probe frame as

$$^p\boldsymbol{p}(i,j) = {}^p\boldsymbol{T}_c\, {}^c\boldsymbol{p}(i,j). \tag{61}$$

The minimum distance of all 3-D pixel coordinates to the probe frame is given by

$$d_o = \min d_p(i,j) \quad i:1,\ldots,W \ \ j:1,\ldots,H \tag{62}$$

where $d_p(i,j) = \|^p\boldsymbol{p}(i,j)\|$ is the Euclidean distance of $^p\boldsymbol{p}(i,j)$ to the probe frame.

Taking into account d_o, the new force variable $\boldsymbol{f}_m \in \mathbb{R}^3$, is given by

$$\boldsymbol{f}_m = \begin{cases} \sigma_d(c, -(d_o - d_t))\boldsymbol{f}_{\text{fs}}, & \text{if } p_c = 0 \\ \boldsymbol{f}_{\text{fs}}, & \text{if } p_c = 1 \end{cases} \tag{63}$$

where $\boldsymbol{f}_{\text{fs}} \in \mathbb{R}^3$ is the force sensor measurement and p_c is a boolean variable used to identify contact

$$p_c = \begin{cases} 1, & \text{if } d_o < d_t \text{ and } f_{\min} < \|\boldsymbol{f}_{\text{fs}}\| \\ 1, & \text{if } d_o > d_t \text{ and } f_{\max} < \|\boldsymbol{f}_{\text{fs}}\| \\ 0, & \text{otherwise} \end{cases} \tag{64}$$

where $f_{\min}$ is a threshold value for inertial forces near contact and $f_{\max}$ is a threshold value for contact not identified by the 3-D camera. σ_d is a general sigmoid function, given by

$$\sigma_d(c,x) = \frac{1}{1+e^{-cx}} \tag{65}$$

which acts as a smooth switch of d_o centered around d_t. The parameter c is a smoothing factor that defines the adaptation dynamics while d_t acts as a threshold for force measurements, anticipating contact.

V. Orientation Control

The orientation control has a joint space velocity control driven by task-space posture errors.

A. Task-Space Orientation Control

Let $\boldsymbol{\omega}_s \in \mathbb{R}^3$ be the angular velocity reference, computed from the orientation error as

$$\boldsymbol{\omega}_s = \boldsymbol{I}\boldsymbol{\epsilon}_{\text{dr}} \tag{66}$$

where $\boldsymbol{I} \in \mathbb{R}^{3\times3}$ is the identity matrix and $\boldsymbol{\epsilon}_{\text{dr}} \in \mathbb{R}^3$ is the orientation error. Let $\mathcal{Q}_d = \{\eta_d, \boldsymbol{\epsilon}_d\}$ and $\mathcal{Q}_r = \{\eta_r, \boldsymbol{\epsilon}_r\}$ be, respectively, desired and current robot orientation expressed in the unit quaternion framework [55]. η and $\boldsymbol{\epsilon}$ are, respectively, the scalar and vector part of the unit quaternion. The mutual orientation between $\mathcal{Q}_d$ and $\mathcal{Q}_r$, expressed in the robot frame $^r\mathcal{Q}_{\text{dr}} = \{\eta_{\text{dr}}, {}^r\boldsymbol{\epsilon}_{\text{dr}}\}$ is given by

$$^r\mathcal{Q}_{\text{dr}} = \mathcal{Q}_r^{-1} * \mathcal{Q}_d. \tag{67}$$

Mapping (67) into the base frame

$$\begin{aligned} \mathcal{Q}_{\text{dr}} &= \mathcal{Q}_r * {}^r\mathcal{Q}_{\text{dr}} * \mathcal{Q}_r^{-1} \\ &= \mathcal{Q}_d * \mathcal{Q}_r^{-1} \end{aligned} \tag{68}$$

while recognizing that $\mathcal{Q}_{\text{dr}} = \{1, \boldsymbol{0}\}$ only if both desired and robot frame are aligned, a minimal orientation error representation can be defined as [56]

$$\boldsymbol{\epsilon}_{\text{dr}} = \eta_r\boldsymbol{\epsilon}_d - \eta_d\boldsymbol{\epsilon}_r - \boldsymbol{S}(\boldsymbol{\epsilon}_d)\boldsymbol{\epsilon}_r \tag{69}$$

where $\boldsymbol{S}(.)$ is the skew-symmetric operator. $\boldsymbol{\omega}_s$ is mapped to the joint space by the inverse differential kinematics equation

$$\dot{\boldsymbol{q}}_s = \boldsymbol{J}_s^{\dagger}(\boldsymbol{q})\boldsymbol{\omega}_s \tag{70}$$

where $\boldsymbol{J}_s(\boldsymbol{q}) \in \mathbb{R}^{3\times n}$ is the orientation Jacobian matrix and † indicates the Moore–Penrose pseudoinverse. $\dot{\boldsymbol{q}}_s$ is the joint space counterpart of $\boldsymbol{\omega}_s$, used to compute the reference for the joint velocity controller.

B. Joint Space Velocity Control

The joint velocity reference $\dot{\boldsymbol{q}}_d$ is computed by scaling $\dot{\boldsymbol{q}}_s$ by a proportional control gain $\boldsymbol{K}_1$

$$\dot{\boldsymbol{q}}_d = \boldsymbol{K}_1\dot{\boldsymbol{q}}_s. \tag{71}$$

In the sequel, the control analysis will be done for each joint individually, being controlled by an AOB.

1) System Plant: The orientation controller is designed assuming the dynamics of a double integrator in the null space [see (35)]. When controlling the system at velocity level, the system plant reduces to a single integrator, given by

$$G(s) = \frac{e^{-sT_d}}{s} \tag{72}$$

where T_d is the system delay. The equivalent temporal representation is given by

$$\dot{y}(t) = \nu(t - T_d) \tag{73}$$

where ν is the plant input (joint acceleration) and $y(t)$ is the plant output (joint velocity $\dot{q}$). Defining $x_{r1}(t) = y(t)$, the state-space representation of (73) is given by (48), with

$$A = 0 \quad B = 1 \text{ and } C = 1.$$

The equivalent discrete system is achieved following the formalism presented in Section IV-A. The AOB algorithm has

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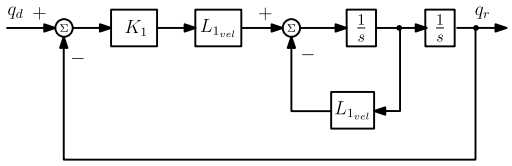


Fig. 3. Simplified orientation control architecture at joint level. The joint control gains are chosen based on this assumption. q_d and q_r are, respectively, desired and robot joint positions.

been discussed in Section IV-B and the orientation control variable ν is given by

$$\nu = L_{1_{\text{vel}}}\dot{q}_d - [\boldsymbol{L}_{r_{\text{vel}}} \quad 1]\begin{bmatrix}\hat{\boldsymbol{x}}_{r,k_{\text{vel}}} \\ \hat{p}_{k_{\text{vel}}}\end{bmatrix}. \tag{74}$$

C. Control Design

The motion control is designed to provide a high tracking performance. $\boldsymbol{L}_{r_{\text{vel}}}$ is computed by the Ackermann's formula [52], with velocity loop time constant τ_v. Setting a small τ_v leads to a fast control response. K_1 is designed assuming that $\boldsymbol{J}_s^{\dagger}(\boldsymbol{q})$ and T_d have no influence in motion control dynamics. Based on this assumption, the pole placement gain reduces to $L_{1_{\text{vel}}}$ and for the desired plant model ($p_{k_{\text{vel}}} = \hat{p}_{k_{\text{vel}}}$), the orientation control closed-loop transfer function is equal to

$$G(s) = \frac{q_r(s)}{q_d(s)} = \frac{K_1 L_{1_{\text{vel}}}}{s^2 + L_{1_{\text{vel}}}s + K_1 L_{1_{\text{vel}}}}. \tag{75}$$

Fig. 3 shows the simplified control architecture. For a critically damped behavior ($\zeta = 1$), K_1 is given by

$$K_1 = \frac{L_{1_{\text{vel}}}}{2^2\zeta^2}. \tag{76}$$

With the presence of $\boldsymbol{J}_s^{\dagger}(\boldsymbol{q})$ in the control law, the decoupled system is no longer achievable. Although being an approximation, this approach provides a "good guess" for the K_1 design, giving a good tracking performance without any notorious overshoots in the end-effector orientation.

D. Inverse Differential Kinematics

Equation (70) computes the joint velocity reference without any force control constraint. As a result, coupling effects with force control torques may arise, being canceled by the null-space projector [see (21)]. Thus, only redundant joints are used for orientation motion, which might introduce orientation errors. Since $\boldsymbol{J}_s^{\dagger}(\boldsymbol{q})$ is the Moore–Penrose pseudoinverse, the solution of (70) will instantly minimize the joint velocity norm and, as long as $\dot{\boldsymbol{q}}_s$ is not parallel to $\boldsymbol{\tau}_p$, orientation motion is going to be performed by redundant joints. As a consequence of this approach, error convergences toward zero might be slower and, eventually, might converge to a nonzero error state if $\dot{\boldsymbol{q}}_s$ is parallel to $\boldsymbol{\tau}_p$. Slower convergence can be mitigated by setting higher control gains.

VI. Online Stiffness Estimation

While an accurate environment description is desirable for contact computer simulation [57] or environment perception [58], from a control standpoint, rather than aiming to achieve a perfect model, the purpose is to obtain an estimation that improves stability and boosts force tracking performance.

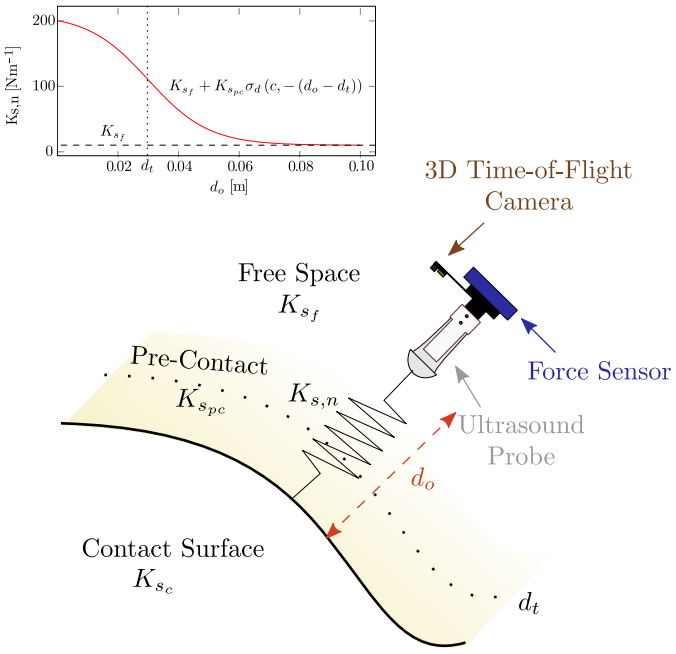


Fig. 4. $K_{s,n}$ adaptation toward contact. K_{s_f}, $K_{s_{\text{pc}}}$, and K_{s_c} are free space, precontact, and contact stiffness parameters, respectively. d_o is the distance to contact given by the depth camera and d_t is a distance threshold.

A. Stiffness Adaptation Toward Contact

Let the environment be modeled as a linear stiffness with a single contact point. Knowing the distance to contact d_o [see (62)], the robot dynamic behavior can be adapted by updating $K_{s,n}$ as

$$K_{s,n} = \begin{cases} K_{s_f} + K_{s_{\text{pc}}}\sigma_d(c, -(d_o - d_t)), & \text{if } p_c = 0 \\ K_{s_c}, & \text{if } p_c = 1. \end{cases} \tag{77}$$

K_{s_f}, $K_{s_{\text{pc}}}$, and K_{s_c} are free space, precontact, and contact stiffness parameters, respectively. K_{s_f} drives the robot dynamics in free space, having a nominal low value so motion control performance is not affected by modeling free space as a spring. It can have a low value since inertial forces in free space are discarded by (63). $K_{s_{\text{pc}}}$ is a precontact stiffness, used to reduce robot dynamics toward contact. It increases system stability when inertial forces are introduced in the control loop (see Section IV-C), allowing a smooth transition from free space motion to contact. K_{s_c} is the contact stiffness. Fig. 4 shows the proposed approach for stiffness adaptation when moving from free space to contact.

B. Contact Stiffness Estimation

In unstructured environments, as the one in ultrasound examination, where free space motion interchanges with soft and stiff contact scenarios, a precise geometrical description of the environment is unfeasible, even when d_o is available. Following the idea introduced in [8] and [46], the relation between desired, measured and estimated forces are used to estimate $\hat{K}_{s_c}$, reducing the mismatch between the environment stiffness K_s and the modeled one $K_{s,n}$. Let f_d, f_m and f_e be desired, measured and estimated forces along one Cartesian direction, with f_e given by the AOB first state estimate. When $K_s \gg K_{s,n}$, the controller anticipates the environment to be

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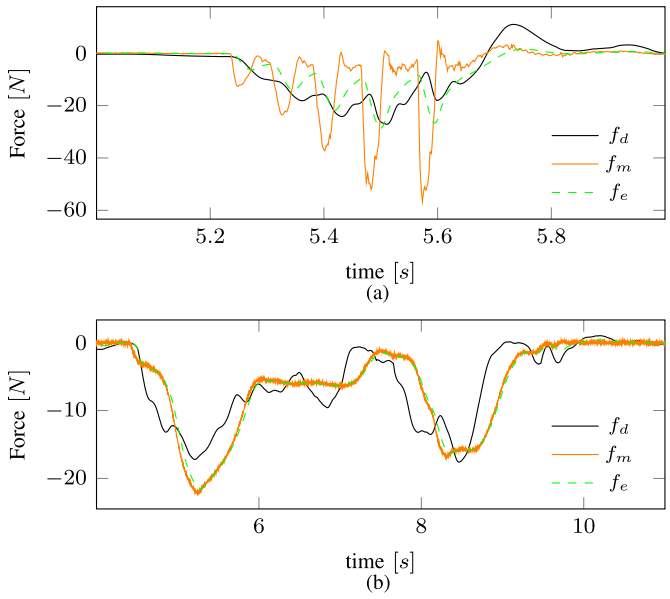


Fig. 5. Force control without stiffness adaptation. (a) Underestimated scenario. $K_{s,n} = 300$ N/m and K_s changes from free space to contact with a hard cover book ($K_s \approx 2000$ N/m). (b) Overestimated scenario. $K_{s,n} = 6000$ N/m and K_s changes from free space to contact with a sponge ($K_s \approx 500$ N/m).

more compliant than it actually is. A low $K_{s,n}$ means high feedback gains, which leads to an overcompensation of force errors, leading to an underdamped response. In this situation, f_m presents an oscillatory behavior which f_e is unable to track, averaging out f_m oscillations. As a consequence, the difference between f_d and f_e is relatively small when compared with the difference between f_m and f_e. Fig. 5(a) shows this behavior. On the other hand, when $K_s \ll K_{s,n}$ [see Fig. 5(b)], the environment is more compliant than expected by the controller, leading to a sluggish response since control gains are too low to overcome force errors. The force estimation is able to track measurements, however, the low control gains lead to a poor f_d tracking, resulting in a small difference between f_m and f_e when compared with the difference between f_d and f_e. Based on this analysis, $\hat{K}_{s,1}$ is estimated by

$$\hat{K}^i_{s,1} = \hat{K}^{i-1}_{s_c} + \Delta\hat{K}^i_{s_c} \tag{78}$$

where

$$\Delta\hat{K}^i_{s,1} = \hat{K}_{s,u} + \hat{K}_{s,o} \tag{79}$$

$$\hat{K}_{s,u} = k_u|f_m - f_e|\sigma_d(c_u, |f_m - f_e| - b_u) \tag{80}$$

$$\hat{K}_{s,o} = -k_o|f_d - f_e|\sigma_d(c_o, |f_d - f_e| - b_o) \tag{81}$$

and $\hat{K}^i_{s,1} \geq 0$. $\hat{K}_{s,u}$ corrects the errors in an underestimated stiffness scenario while $\hat{K}_{s,o}$ compensates for overestimated ones. System stability is guaranteed by the estimation algorithm design. Whenever the system presents an underdamped response, the stiffness is increased reducing the control feedback gains and stabilizing the system. σ_d is given by (65) and b_u, b_o, c_u, c_o, k_u, and k_o are all positive parameters.

In [8], a second term increasing stiffness with contact force is also included; however, in [8], the end-effector orientation is kept constant. When performing orientation motions, the inertial properties perceived at the end-effector change with the manipulator posture [59], changing the contact dynamics. Interestingly, we noted that, rather than changing with the contact force, the stiffness perceived by the controller changes with the effective mass referred to the end-effector. The perceived stiffness can be anticipated by the effective mass as

$$\hat{K}^i_{s,2} = k_m\, m_j \tag{82}$$

where k_m is a positive gain and m_j is the effective mass along the jth Cartesian direction, given by the jth main diagonal element of $\boldsymbol{\Lambda}_p(\boldsymbol{q})$ [see (9)]. Therefore, we propose to estimate the contact stiffness as

$$\hat{K}_{s_c} = \hat{K}^i_{s,1} + \hat{K}^i_{s,2}. \tag{83}$$

To avoid jerky behaviors from a too low $\hat{K}_{s_c}$ estimation, K_{s_c} in (77) is given by

$$K_{s_c} = \begin{cases} \hat{K}_{s_c}, & \text{if } \hat{K}_{s_c} \geq K_{s_{c,\min}} \\ K_{s_{c,\min}}, & \text{if } \hat{K}_{s_c} < K_{s_{c,\min}} \end{cases} \tag{84}$$

where $K_{s_{c,\min}}$ is the minimum contact stiffness to limit control gains in contact. It is given by

$$K_{s_{c,\min}} = K_{s_f} + K_{s_{\text{pc}}}. \tag{85}$$

Low-pass filters are used in $K_{s,n}$ and in d_o computation, with the filter equations given by

$$K^{f,i}_{s,n} = K^{f,i-1}_{s,n} + l_{s,n}\left(K^i_{s,n} - K^{f,i-1}_{s,n}\right) \tag{86}$$

$$d^{f,i}_o = d^{f,i-1}_o + l_o\left(d^i_o - d^{f,i-1}_o\right) \tag{87}$$

where the superscript i indicates the iteration and f indicates the filtered value.

C. Virtual Coupling Computation

The relation between K_{virt} and $K_{s,n}$ is crucial to provide good position and force tracking. We propose to compute K_{virt} as

$$K_{\text{virt}} = \begin{cases} K_{\text{virt}_f} + k_{v_{\text{pc}}} K_{s_{\text{pc}}} \sigma_d(c, -(d_o - d_t)), & \text{if } p_c = 0 \\ K_{\text{virt}_f} + k_{v_{\text{pc}}} K_{s_{\text{pc}}} + \hat{K}_{s,2}, & \text{if } p_c = 1 \end{cases} \tag{88}$$

where K_{virt_f} is the virtual coupling value in free space. The second term in (88) increases K_{virt} proportionally to $K_{s,n}$ in precontact situations, with $k_{v_{\text{pc}}}$ given by the ratio between $K_{\text{virt}_f}/K_{s_f}$. In contact, $\hat{K}_{s,2}$ drives K_{virt} (see Section VI-B) with $K_{\text{virt}} \leq K_{\text{virt}_M}$. K_{virt_M} is a threshold value to limit K_{virt} maximum value.

VII. Experiments

The goal of the experimental section is to validate and assess the proposed control architecture in multiple contact/noncontact situations. Our goal is to analyze force and motion control performance in several telemanipulation scenarios, ranging from "academic" (such as interacting with a stiff object) to clinical ones, where a physician performs a teleechographic exam on a healthy volunteer. The experiments

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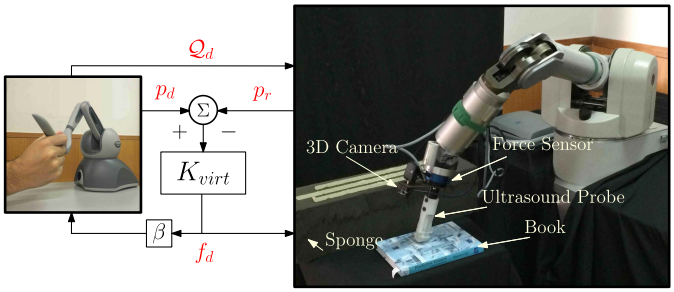


Fig. 6. Teleoperation scheme and experimental setup used in the first set of experiments. β is a force scaling factor. Throughout the experiments $\beta = 0.1$.

show all major and innovative aspects of the control architecture, namely: 1) online stiffness estimation and its relation with the robot effective mass and 2) motion and force tracking performances in the above mentioned scenarios.

Clinical and nonclinical experiments are performed to assess both the stiffness estimator and the control architecture. In the nonclinical one, the relation between effective mass and stiffness perceived by the controller is investigated. In the clinical experiment, a physician performs a robotic-assisted tele-echography examination on a healthy volunteer to assess system performance.

A. Experimental Setup

The medical robot used in the experiments is a torque controlled 7-DoF WAM lightweight anthropomorphic robot. The manipulator is controlled through a Phantom Omni haptic device, being both devices connected through a local area network. A JR3 force sensor is attached to the end-effector. In the force sensor, an ultrasound probe and a 3-D time-of-flight camera are also attached. The ultrasound probe is a SonoSite Titan C15 curved transducer while the 3-D camera is a PMD[vision] CamBoard nano. Fig. 6 shows the experimental apparatus for the first experiment.

B. Control Design

The control sampling time and the system time delay are $h = 1.2$ ms and $T_d = h$, respectively.

1) Force Control Design: Typically, the manipulator structural damping K_2^s is neglected ($K_2^s = 0$ and $K_2^t = K_2^v$). In our setup, setting $K_2^t = K_2^v = 16.6$ provides a good control performance, however, it introduces a noticeable high-frequency noise perturbation from a too a high K_2^v. Reducing both K_2^t and K_2^v affects the control performance, decreasing the system responsiveness. Recognizing that the manipulator structure has some internal damping ($K_2^s > 0$), K_2^t in the AOB model can be higher than the velocity loop feedback gain K_2^v [see (46)], reducing the high-frequency noise without any notorious drop off in the control performance. Following this approach, the damping parameters are set at $K_2^t = 16.6$ and $K_2^v = 12.5$, being both parameters obtained experimentally. The force control time constant is set at $\tau_f = 0.035$ s.

Table I gives the parameters for $\hat{K}_{s_c}$ estimation. Since the underdamped response is critical for system stability,

TABLE I
ONLINE STIFFNESS ESTIMATION DESIGN PARAMETERS

b_u	0.7	c_u	9.0	k_u	7.0	$[m^{-1}]$
b_o	1.0	c_o	7.0	k_o	3.0	$[m^{-1}]$
				k_m	500	$[NKg^{-1}m^{-1}]$

TABLE II
NUMERICAL VALUES OF $K_{s,n}$ AND K_{virt} DESIGN PARAMETERS

c	100	d_t	0.03	$[m]$	K_{s_f}	10	$[Nm^{-1}]$
$k_{v_{pc}}$	4.0	f_{min}	2.0	$[N]$	$K_{s_{pc}}$	200	$[Nm^{-1}]$
$l_{s,n}$	0.05	f_{max}	4.0	$[N]$	K_{virt_f}	40	$[Nm^{-1}]$
l_o	0.05				K_{virt_M}	2000	$[Nm^{-1}]$

the estimation algorithm is designed to react faster in stiffness underestimated scenarios than in overestimated ones ($k_u > k_o$, $c_u > c_o$, and $b_u < b_o$). In underestimate scenarios, control gains are quickly reduced, keeping system stability, while in overestimated ones, the stiffness decreases more slowly. Table II gives the parameters for K_{virt} and $K_{s,n}$ computation. The null-space gain $\boldsymbol{K}_N \in \mathbb{R}^{7\times 7}$ is

$$\boldsymbol{K}_N = \text{diag}[50 \quad 50 \quad \dots \quad 50].$$

In free space the pole placement gains are

$$\boldsymbol{L}_r = [79.68 \quad 3.98 \quad 0.048]$$

being adapted according to (60) when $K_{s,n}$ change. For the AOB stochastic design, a sensor-based strategy (low $\boldsymbol{Q}_k$ when compared to $\boldsymbol{R}_k$) is pursued with

$$\boldsymbol{Q}_k = \text{diag}[10^{-1} \quad 10^{-12} \quad 10^{-12} \quad 3\times 10^{-2}]$$

and $\boldsymbol{R}_k = 1$, leading to the following steady-state Kalman gain in free space

$$\boldsymbol{K}_k = [0.2705 \quad 0.1133 \quad 0.148 \quad 0.148]^T.$$

Again, when $K_{s,n}$ changes, $\boldsymbol{K}_k$ also has to be updated. Due to force and position measurements noise, $\boldsymbol{f}_d$ and $\boldsymbol{f}_m$ are both filtered with a first-order low-pass Butterworth filter with cutoff frequencies 25 and 5 Hz, respectively. Prefiltering force signals allow setting higher control gains and a higher sensor-based stochastic design, which is adequate for tele-echography tasks. A sensor-based design enables better free space position tracking, reducing the effect of model errors introduced by modeling free space as a spring, which becomes notorious in precontact scenarios when the model stiffness is increased by (77). In contact, a sensor-based design keeps the system responsive and, together with the contact anticipation strategy, avoids jerky behaviors when changing from free space to stiff contact scenarios.

2) Motion Control Design: Table III summarizes motion control gains.

For the AOB stochastic design, all joints present the same sensor-based approach

$$\boldsymbol{Q}_{k_{\text{vel}}} = \text{diag}[10^{-1} \quad 10^{-12} \quad 10^{-12}]$$

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TABLE III
MOTION CONTROL DESIGN

	τ_v	K_1	$L_{1_{vel}}$	$L_{2_{vel}}$
q_1	0.008	29.02	116.08	0.14
q_2	0.008	29.02	116.08	0.14
q_3	0.0075	30.8	123.21	0.15
q_4	0.0075	30.8	123.21	0.15
q_5	0.00625	36.39	145.58	0.17
q_6	0.00625	36.39	145.58	0.17
q_7	0.01	23.56	94.23	0.11

entailing the following steady-state Kalman gain:

$$\boldsymbol{K}_{k_{\text{vel}}} = [0.2702 \quad 3.24 \times 10^{-10} \quad 3.24 \times 10^{-10}]^T.$$

High control gains and a sensor-based strategy address the orientation control.

3) 3-D Camera Design: The camera integration time is set at 1000 μs and the amplitude threshold is set to 4000. This setup provides good environment description without probe detection. We have used the manufacturer calibration data.

C. Experimental Validation of Stiffness Estimation

Two experiments were conducted to assess the stiffness estimation algorithm. In the first experiment, the relation between the inertial properties perceived at the end-effector and the contact stiffness K_{s_c} perceived by the controller is emphasized, while in the second experiment, the estimation algorithm performance is compared with the approach presented in [8] in two distinct contact postures, where the inertial properties are differently perceived.

The first experiment can be summarized as follows. The probe motion starts in free space with a vertical posture, reaching contact with a stiff surface [hard cover book, see Fig. 7(a)] around 12 s. While in contact, the probe rotates along a perpendicular axis, from its initial contact posture to an horizontal one [20–30 s, see Fig. 7(b)] and then back to its initial orientation (35–50 s). $\hat{K}_{s_c}$ is only estimated by force relations [see (78)], and the reference force is kept constant during the experiment. The overall motion is performed with slow dynamics to minimize disturbances in force control and consequently, in the stiffness estimation, highlighting the relation between perceived stiffness and effective mass.

The second experiment can be described in the following way. The probe motion starts in free space, reaching contact with the same stiff surface two times, first with a vertical posture (high effective mass) [8–18 s, Fig. 7(a)] and then with an horizontal one [low effective mass, 26–45 s, Fig. 7(b)]. The probe posture and the reference force are kept constant in both contact scenarios. Between contacts, the orientation is changed in free space to the second contact posture. The experiment is performed two times, first using the stiffness estimation approach proposed in [8] and then repeated using the estimation algorithm presented in this paper. The second term of (83) is computed in [8] by

$$\hat{K}^i_{s,2} = k_3 \sigma_d (c_0, |f_m| - f_0) \tag{89}$$

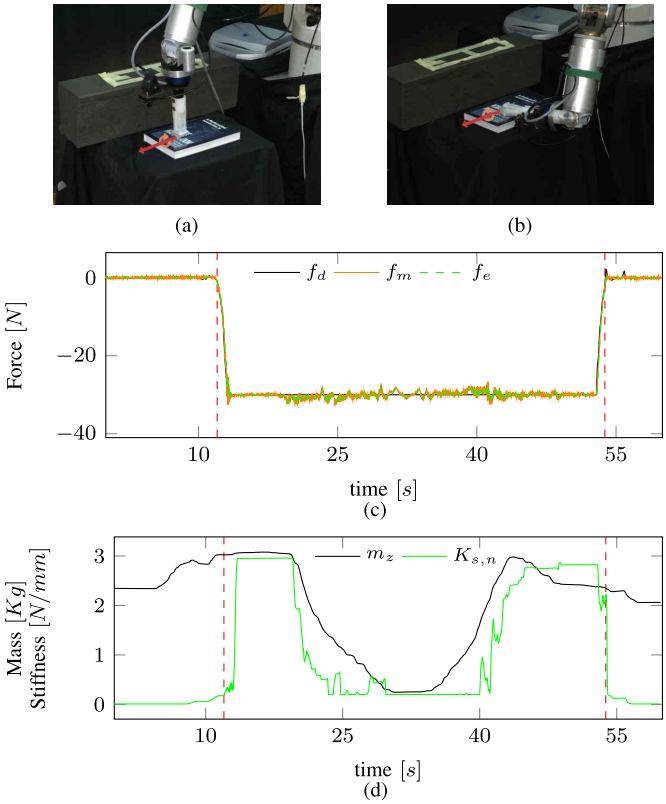


Fig. 7. Stiffness perception by the controller when the end-effector posture changes constrained by a stiff surface (book). (a) and (b) Probe at vertical and horizontal postures, respectively. (c) Force tracking in Z Cartesian dimension with online stiffness estimation [only by (78)] and control adaptation. f_d, f_m, and f_e are desired, measured, and estimated forces, respectively. (d) Effective mass m_z and stiffness estimation $K_{s,n}$ in Z Cartesian dimension. To highlight the connection between both variables, $K_{s,n}$ is plotted in newton per millimeter. Desired force and probe position are kept constant through the constrained orientation motion. Red dashed vertical lines identify free space to contact and vice-versa transitions.

increasing the stiffness estimation with the measured force f_m. $k_3 = 3000$ Nm^{-1}, $c_0 = 0.2$, and $f_0 = 20$ N are set according to [8]. σ_d is given by (65). In both experiments the contact surface is placed parallel to the robot base.

1) Discussion: Fig. 7 shows the result of the first experiment. The reference force is well tracked, validating the stiffness estimation throughout the experiment [see Fig. 7(c)]. From Fig. 7(d), a connection between the effective mass and the perceived stiffness can be inferred. As can be seen, in contact, excluding the transitory behavior from free space to contact, the stiffness estimation and the effective mass have a similar curve (13–54 s), while the contact force seems to have no effect in the stiffness estimation. Clearly, the physical properties of the book do not change throughout the experiment, however, from a control standpoint, the environment is differently perceived when the end-effector posture changes. The effective mass can be used to anticipate the perceived stiffness [see (82)], improving system performance in dynamic interactions.

Fig. 8 shows the results from the second experiment. When the stiffness is estimated by [8], the system displays two distinct behaviors, according to how the effective mass is

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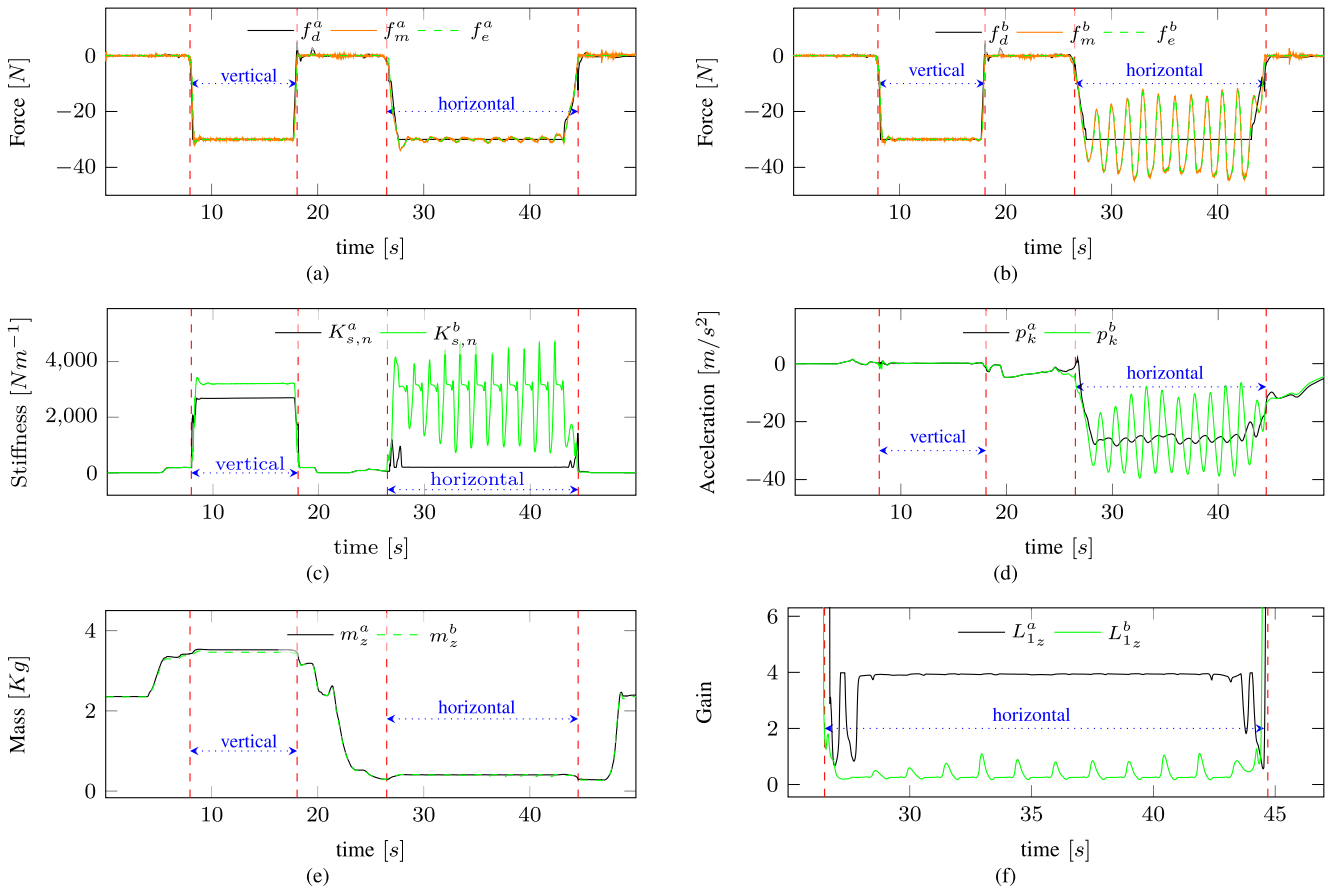


Fig. 8. Experimental results of the second experiment. The performance of the stiffness estimator (identified by the superscript ^{a}) is compared with the estimator presented in [8] (superscript ^{b}) in two distinct end-effector contact postures (vertical and horizontal contact postures). (a) and (b) Force tracking in Z Cartesian dimension with online stiffness estimation and control adaptation. f_d, f_m, and f_e are desired, measured, and estimated forces, respectively. (c) Stiffness estimation $K_{s,n}$ through the experiment. (d) AOB active state p_k evolution. (e) Effective mass perception at end-effector along the Z Cartesian dimension. (f) First element of Z Cartesian dimension state feedback gain L_{1_z} in the second contact. Red dashed vertical lines identify free space to contact and vice-versa transitions, while blue dotted horizontal lines identify the end-effector contact posture.

perceived at end-effector. When the effective mass is high [Fig. 8(e)], f_d is effectively tracked, validating the stiffness estimation. However, when the effective mass is low, f_m oscillates around the reference force, leading to a poor f_d tracking performance. The relation between f_m, f_e, and f_d in the second contact, shows a typical stiffness overestimation behavior (low control gains), with f_e accurately estimating f_m, but unable to track f_d [Fig. 8(b)]. Due to low control gains [see Fig. 8(f)], most of the control effort comes from the AOB active state p_k [see Fig. 8(d)], which is not enough to tackle the overestimation disturbance, leading to f_m oscillatory behavior. Since $\hat{K}_{s_c}$ is driven by f_m [see (89)], its oscillatory behavior leads to the stiffness oscillations shown in Fig. 8(c).

On the other hand, when using the approach presented in this paper, the reference force is well tracked in both contact postures [Fig. 8(a)], validating the estimation approach. The environment is differently perceived by the controller [Fig. 8(c)], despite the constant contact force, which shows that f_m absolute value *per se* has no influence in $\hat{K}_{s_c}$ estimation [see (89)]. In the second contact, excluding the transitory behavior from free space to contact and vice versa, the estimation algorithm keeps the stiffness value constant at $K_{s_{c,\min}}$, stressing that to track f_d when the effective mass is low, high control gains are required, i.e., when the effective mass changes, also the contact dynamics and the controller perception of the environment changes.

Furthermore, the AOB active state evolution [Fig. 8(d)] shows that, while in the first contact the active state has little influence in the control command, in the second contact, a significant amount of the control effort comes from p_k, meaning that the environment model is less precise and the model error is tackled by the AOB active state. Another important remark is that, from a control standpoint, a precise estimation is not critical for a good system performance, since in the first contact, f_d is effectively tracked with both estimation algorithms despite the estimation numerical value be somewhat different (around 500 Nm^{-1}).

D. Robotic-Assisted Tele-Echography Examination

To validate the control architecture in a clinical context, a gynecologist performed a robotic-assisted pelvic ultrasound examination in a healthy volunteer. The physician tele-operates the probe holder manipulator through a Phantom Omni haptic device, sending 6-DoF (position and orientation) commands, and receiving 3-DoF force feedback. The examination is performed with the patient urinary bladder full, which acts as

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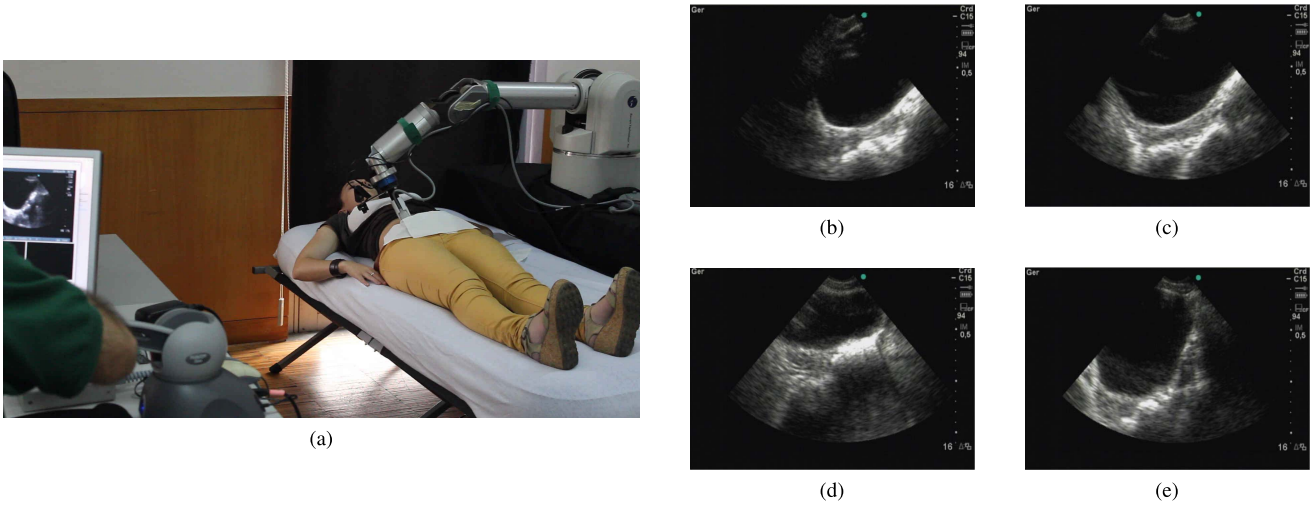


Fig. 9. Robotic-assisted pelvic ultrasound examination. (a) Physician performing the robotic-assisted ultrasound examination in a healthy volunteer. (b)–(e) Ultrasound images acquired by the physician. (b) Uterus cross section. (c) Uterus midsagittal section. (d) Left ovary. (e) Right ovary.

an acoustic window for the pelvic organs, namely, the urinary bladder itself, the uterus and the ovaries. Fig. 9(a) shows the physician performing the ultrasound procedure while Fig. 9(a)–(e) shows ultrasound images acquired in the pelvic examination.

The procedure starts with the probe in free space, reaching contact with the patient abdominal region around 12 s. Contact is kept during the examination (12–111 s), except at 17 s, when the physician repositions the probe in free space and at the end of the procedure (around 109 s).

1) Discussion: The pelvic ultrasound examination was chosen for two reasons. First, it is a dynamic procedure, requiring the probe to be placed in different locations with different postures, which highlights the system robustness and the control performance. In addition, the effectiveness of the robotic-assisted examination can be extrapolated to less dynamic abdominal procedures, validating the control architecture in those procedures. Second, it is easily performed in our lab without requiring to change the manipulator working space envelope (and hence its base) to a new position. From a clinical perspective, the physician performed the pelvic examination effortlessly, being able to identify different anatomical regions from the haptic feedback. From a control standpoint, force and position are well tracked, as can be seen from Figs. 10 and 11, respectively. Free space to contact and vice-versa transitions present no jerky behaviors, there are no noticeable steady-state errors in force tracking and the position tracking in contact is consistent with soft tissue interaction. Fig. 12 shows the orientation control performance. In general, the orientation is well tracked, despite a few tracking errors arising due to the hierarchical control strategy, as explained in Section V-D. These errors can be tackled with higher control gains, however, due to the task nature (interaction with humans), a more conservative control design is followed. Nevertheless, the physician ability to perform the examination is not affected, validating the hierarchical approach to compute orientation motion. Finally, Fig. 13 shows $K_{s,n}$ and K_{virt} computation. A good estimation of contact stiffness can be inferred, since f_m tracks f_d with high performance

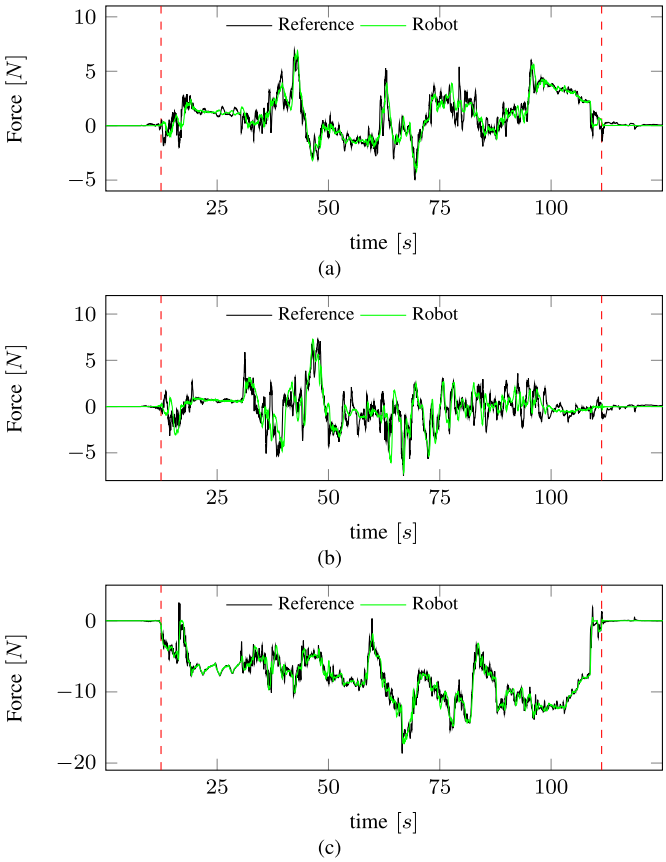


Fig. 10. Force tracking with stiffness adaptation in the robotic-assisted ultrasound examination. (a)–(c) Cartesian force data in X, Y, and Z, respectively. Red dashed vertical lines identify free space to contact and vice-versa transitions.

(see Fig. 10). Table IV shows root-mean-square error (RMSE) and absolute mean error (AME) for force tracking. Low tracking errors validate the perceived stiffness estimation.

VIII. Discussion

This paper presents a novel control architecture for robotic teleoperation. The control architecture relies on computed-torque techniques that take into account robot dynamics,

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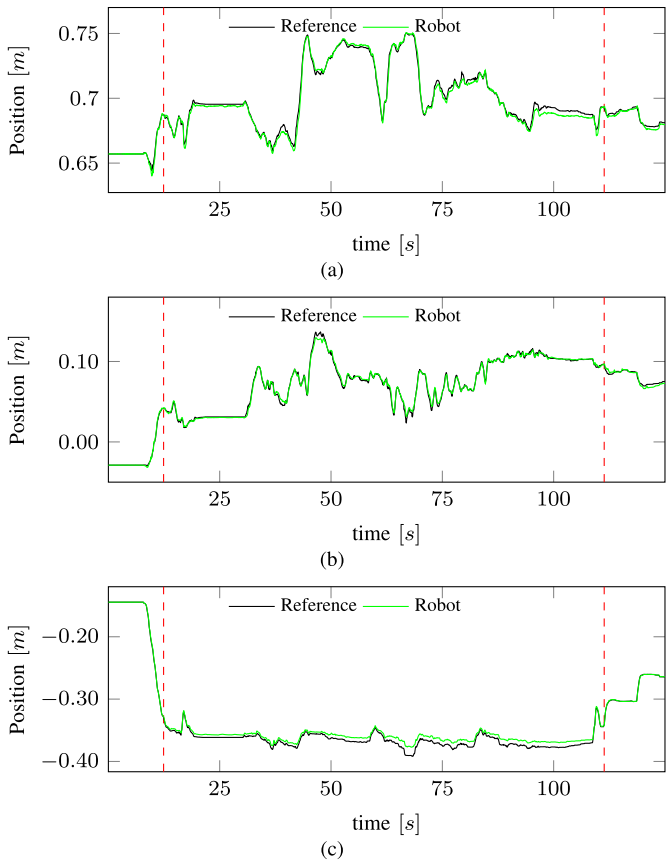


Fig. 11. Cartesian position tracking in the robotic-assisted ultrasound examination. (a)–(c) Cartesian position data in X, Y, and Z, respectively. Red dashed vertical lines identify free space to contact and vice-versa transitions.

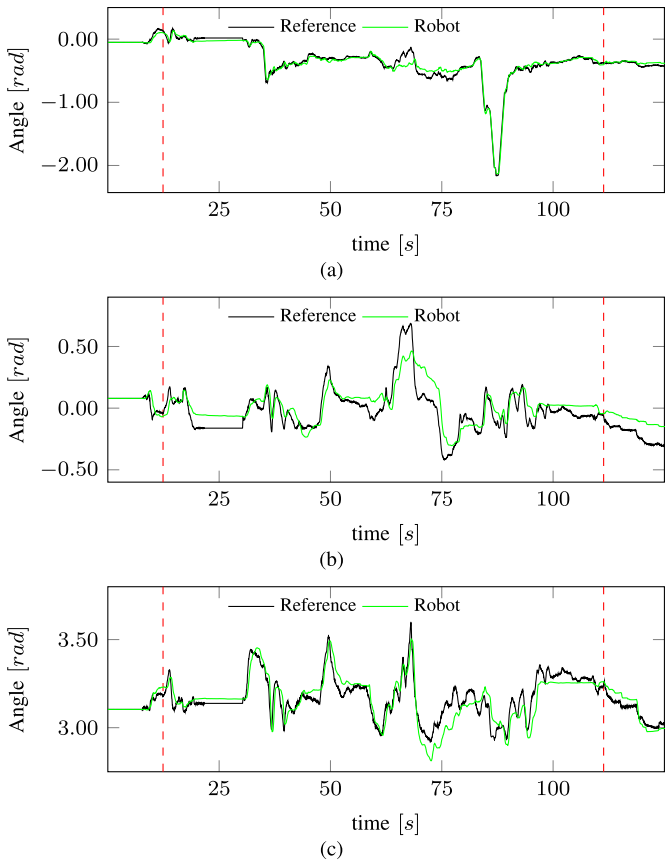


Fig. 12. Orientation tracking in the robotic-assisted ultrasound examination.. The orientation is represented as (a) roll, (b) pitch, and (c) yaw angles. Roll $\in [-\pi, \pi]$, pitch $\in [-\pi, \pi]$, and yaw angle $\in [0, 2\pi]$. Red dashed vertical lines identify free space to contact and vice-versa transitions.

contact stiffness estimation, task-space formulation, and null-space design. A dynamically consistent hierarchical control architecture is proposed, with explicit force control arising as the primary task, while orientation is performed in the null space, reducing coupling effects. An explicit Cartesian force control driven by position errors is regarded as the highest priority task, enforcing a compliant behavior at position level. Probe orientation is controlled at joint level, driven by task-space orientation errors converted into joint velocity references, enforcing a stiff behavior by controlling the orientation at motion level. Controlling the orientation in the joint space allowed achieving a good tracking performance, addressing the limitations found with task-space orientation controllers.

The teleoperation architecture relies on the virtual coupling solution presented in [8] and [46], which greatly simplifies the tele-operation design, allowing to keep the force control architecture in free space. In this way, to improve free space position tracking, the stiffness parameter that enters in the force control loop through the AOB plant model has to be designed with a low nominal value in free space, reducing the detrimental effect of modeling free space as a spring by increasing feedback control gains. When moving toward contact, we propose to anticipate the stiffness parameter using 3-D camera data, allowing adaptation before contact. This procedure decreases robot dynamics toward contact by reducing control gains, allowing a smooth transition from free space to contact. Free space inertial forces that deteriorate control performance are also eliminated from the control loop using depth camera data, improving free space control by providing reliable force measurements.

In contact, the unstructured nature of the echographic procedure requires a controller to cope with different environments. The contact stiffness is estimated resorting only to force data, being updated based on force relations, avoiding geometric uncertainty issues, also following the idea introduced in [8]. The stiffness estimation does not increases with the contact force (as in [8]) but rather changes with the effective mass. This paper shows that different end-effector contact orientations lead to different force responses (the effective mass is configuration dependent). Using the effective mass in the estimation algorithm, the perceived stiffness can be anticipated, improving force control performance in dynamic interactions, specially with stiff surfaces. Clearly, the elastic properties of an object do not change with probe interaction, however, the controller perception of the object does change, existing a strong correlation between the effective mass and the perceived stiffness. In this way, the perceived stiffness can be seen, not as a well defined physical property, but as a force control optimization parameter.

When comparing this architecture with other control architectures for robotic-assisted tele-echography, its major strength is the ability to effectively achieve both force and motion

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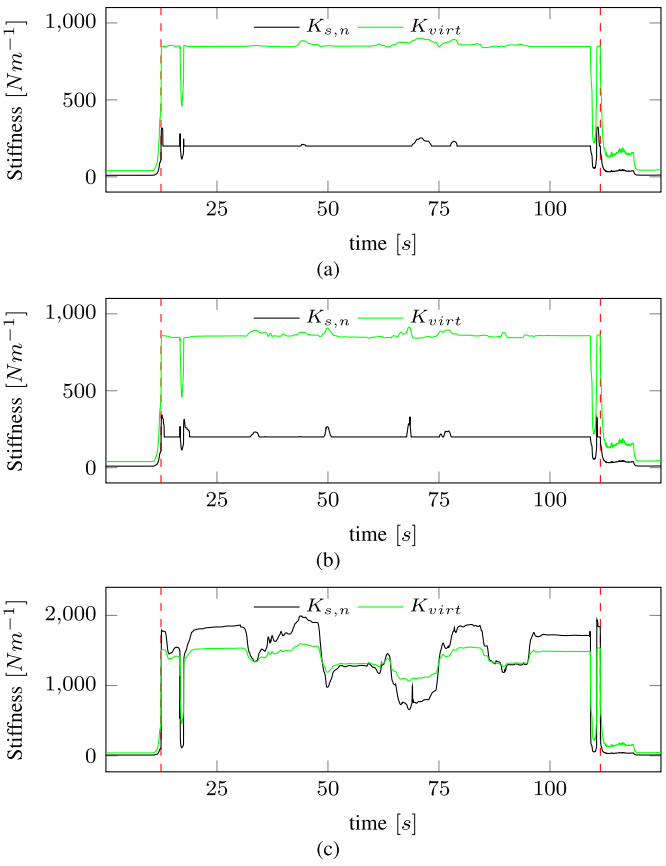


Fig. 13. $K_{s,n}$ and K_{virt} online computation in the robotic-assisted ultrasound examination. (a)–(c) Cartesian stiffness estimations and virtual coupling in X, Y, and Z, respectively. Red dashed vertical lines identify free space to contact and vice-versa transitions.

TABLE IV
AME AND RMSE FOR FORCE TRACKING IN THE CLINICAL EXPERIMENT

$[N]$	AME	RMSE
x	0.403	0.619
y	0.696	1.071
z	0.446	0.718

tracking without control switching. Conti *et al.* [60] presented a control architecture for robotic-assisted tele-echography also based on virtual coupling and force control through AOBs. However, in [60], the AOB is designed with a constant stiffness value, without performing stiffness estimation or adaptation. Although the AOB is robust enough to cope with environment mismatches, to achieve high performance both in free space (position tracking) and contact (force tracking), specially in interaction with different environments (e.g., bone and abdominal contacts), environment estimation and control adaptation are mandatory. This paper tackles these issues in a robust way, based on the innovative control architecture, which includes a new perspective to understand the stiffness perception seen by the controller. In fact, the robot effective mass changes the interaction dynamics, and therefore the contact perception also changes with robot posture.

A. Future Work

1) Robustness Analyses in the Presence of Random Communication Delays: The main focus of this paper has been on the control of the slave manipulator. All teleoperation experiments were performed in a local area network with negligible time delays. When performing a robotic-assisted ultrasound examination between remote locations, delays are always present. If the communication channel is a packet-switched network, such as Internet, the delays are not deterministic, and packets can be lost, which can affect system performance.

2) Estimation of Contact Forces: A force sensor is expensive and a source of noise. De Luca and Mattone [61] presented a residual-based method to detect collisions without using force or acceleration measurements. The simple theoretical formulation of this method is attractive to estimate contact forces. Preliminary work shows potential to provide a reliable estimation of contact forces, allowing to remove explicit force sensing from the control loop.

IX. CONCLUSION

This paper proposes a new control approach for robotic-assisted tele-echography. The control architecture relies on a hierarchical approach, where an explicit Cartesian force control arises as the primary task, establishing a suitable dynamic behavior between probe and patient, while the orientation motion is performed in the null space. The manipulator motion and contact dynamics are driven by a 3-D time-of-flight camera and a force sensor. The aim of this approach is twofold: first, it enables contact anticipation, reducing the robot dynamics before contact; second, it eliminates free space inertial forces from the control loop. In contact, the environment stiffness is estimated online using force data and the manipulator inertial properties referred to the the end-effector. Two sets of experiments have been performed to assess the stiffness estimation algorithm. These experiments illustrate a strong correlation between the stiffness perceived by the controller and the effective mass referred to the end-effector. Taking into account the stiffness estimation in the control loop, accurate force tracking has been achieved, outperforming the results of the estimation algorithm presented in [8]. The control architecture has been assessed in a clinical scenario, with a gynecologist performing a robotic-assisted pelvic ultrasound examination in a healthy volunteer. The physician effortlessly performed the procedure, validating the control architecture in a medical context.

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